Computing internal homs of module category actions

by

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DEDICATION

I would like to dedicate this thesis to my parents. They have stood up for me, and pushed me to challenge myself in my pursuit of mathematics. They have been with me each step of the way, believing in me and guiding me. Without them by my side, I would not have achieved nearly as much, nor be who I am today.

I would also like to dedicate this thesis more broadly to all those family, friends, and mentors who have provided me with the nurturing and the opportunity to accomplish what I have so far. I truly believe that it takes a village, and I have had the privilege and joy to be surrounded by wonderful people who have helped me along the way.

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ABSTRACT

In this thesis, we will be interested in computing internal homs with respect to module category actions. We will first compute a known example for the category of right modules internal to a fusion category. We then will focus on internal homs over equivariantizations of fusion categories, recovering as a special case known results from Etingof and Ostrik (2003). This will stress the importance of "induction functors", which lift equivariant structure. We will generalize the classic induction functors for representations of finite groups, as well as an induction functor found in Drinfeld et al. (2010). This allows us to compute the internal homs in the case of the equivariantization of a fusion category with a group action satisfying a 2-cocycle condition acting on the equivariantization with respect to a subgroup.

CHAPTER 1. INTRODUCTION

The notion of "categorification" has been an avenue with which mathematicians generalize set theoretic objects. For example, monoidal categories seek to generalize the classic notion of a monoid. This replaces the set, on which we have a multiplicative structure, with the objects of a category. Tensor categories, in that regard, are much like a categorification of rings. They have a multiplicative and additive structure on their collection of objects that behave somewhat similarly to rings (with the main difference being the lack of an additive inverse). At the same time, tensor categories are a natural place to study the representation theory of finite groups from a categorical perspective. As we will see in section 2.1, the categories of finite dimensional representations of a finite group naturally arise using equivariantization.

Studying modules over a ring, or representations over a group, have been an invaluable tool in studying the structure of the ring/group itself. Similarly, we may define module categories over a tensor category, which play an important role in studying their structure. Ostrik (2003) introduced the notion of an internal hom of a module category action. These are useful when studying the module categories, since they can allow one to produce an internal algebra whose category of internal right modules is equivalent to the module category one started with. Being able to work with categories of internal right modules proves very helpful when studying module categories.

1.1 Scope of This Work

For the remaining sections of chapter 1, we will review many well known concepts such as monoidal categories, rigid monoidal categories and duals, abelian categories, tensor categories, fusion categories, and module categories. The discussion of these topics in this thesis follows

Etingof et al. (2015), adding commentary when illuminating, and results when needed for the discussions in chapter 2 and chapter 3.

In chapter 2, we review the equivariantization of monoidal/fusion categories, again following the guidance of Etingof et al. (2015). We also discuss equivariant module categories, which were introduced in Etingof et al. (2011), but whose definitions needed to be fixed and modified due to an issue found in the original definitions. We then provide an updated proof sketch to the classification of exact indecomposable module categories over the equivariantization of a fusion category in terms of equivariant module categories.

In chapter 3, we review the notion of an internal hom to a module category action on a fusion category. We derive a known result about the internal homs for module categories of right modules of an algebra internal to a fusion category. We then compute the internal homs of module categories over the category of finite dimensional (linear) representations of a finite group. This result agrees with a special case mentioned in Etingof and Ostrik (2003). From there, we make a modest generalization of this, computing internal homs for equivariant module categories with trivial action over the same fusion category with trivial action. A key result to this is our generalization of Frobenius reciprocity to representations valued in an arbitrary fusion category. After this, we inspect when one tries to consider a fusion category as a module category over itself, and looks at equivariant module structures coming from the action of a finite group. In doing this we generalize the induction functor of Drinfeld et al. (2010) and prove a further generalized Frobenius reciprocity.

We assume the reader is familiar with the basic notions and concepts from category theory; categories, functors, natural transformations, adjunctions, etc. All categories are assumed to be locally small, we denote the hom sets by $\mathrm{Hom}_{\mathcal{C}}(-,-)$. We fix $\Bbbk$, an algebraically closed field of characteristic 0.

1.2 Monoidal Categories

We will not explore the theory of monoidal categories in general, we will merely list the required definitions. This includes defining monoidal categories, monoidal functors between them, and monoidal natural transformations. For greater detail in this subject matter, see MacLane (1978) or Etingof et al. (2015).

Monoidal categories, as briefly stated in the abstract, are categorifications of a monoid. Recall that monoids are sets equipped with a multiplication which is associative and has a multiplicative identity. A monoidal category will then be a category equipped with a "multiplication" on it's objects that is "associative" and has a "multiplicative identity". We find that we need more data than in the normal set case to make this multiplication cohere to the additional categorical structure. Since many objects may lie in the same isomorphism classes, asking for associativity and multiplicative identity on the nose is too strict, so everything is done up to natural isomorphism. We also need to have some coherency conditions to make sure that these choices of natural isomorphisms behave well.

Definition 1.1. A *monoidal category* $\mathcal{C}$ is a sextuple $(\mathcal{C}, \otimes, a, 1, l, r)$ where $\mathcal{C}$ is a category, $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is a bifunctor, $a(X, Y, Z) : (X \otimes Y) \otimes Z \xrightarrow{\sim} X \otimes (Y \otimes Z)$ is a natural isomorphism (called the *associativity constraint*), $1 \in \mathcal{C}$ (called the *identity object*), and natural isomorphisms $l(X) : 1 \otimes X \xrightarrow{\sim} X$, $r(X) : X \otimes 1 \xrightarrow{\sim} X$ (called the *left and right unit constraints*) such that the following axioms hold:

1. The Pentagon Axiom: For $W, X, Y, Z \in \mathcal{C}$, the following commutes

$$\begin{tikzcd}
 & ((W \otimes X) \otimes Y) \otimes Z \arrow[dl, "a(W,X,Y)\otimes \mathrm{id}_Z"'] \arrow[dr, "a(W\otimes X,Y,Z)"] & \\
(W \otimes (Y \otimes X)) \otimes Z \arrow[d, "a(W,X\otimes Y,Z)"] & & (W \otimes X) \otimes (Y \otimes Z) \arrow[d, "a(W,X,Y\otimes Z)"] \\
W \otimes ((X \otimes Y) \otimes Z) \arrow[rr, "\mathrm{id}_W \otimes a(X,Y,Z)"] & & W \otimes (X \otimes (Y \otimes Z))
\end{tikzcd}$$

2. The Triangle Axiom: For $X, Y \in \mathcal{C}$, the following commutes

$$\begin{tikzcd}
(X \otimes 1) \otimes Y \arrow[rr, "a(X{,}1{,}Y)"] \arrow[dr, "r(X)\otimes \mathrm{id}_Y"'] & & X \otimes (1 \otimes Y) \arrow[dl, "\mathrm{id}_X \otimes l(Y)"] \\
& X \otimes Y &
\end{tikzcd}$$

Remark 1.2. Since $\otimes$ is a bifunctor, a common trick we will use in arguments is that for morphisms $f : X \to X'$ and $g : Y \to Y'$ in $\mathcal{C}$, the following commutes:

$$\begin{tikzcd}
X \otimes Y \arrow[r, "f\otimes \mathrm{id}"] \arrow[d, "\mathrm{id}\otimes g"] \arrow[dr, "f\otimes g"] & X' \otimes Y \arrow[d, "\mathrm{id}\otimes g"] \\
X \otimes Y' \arrow[r, "f\otimes \mathrm{id}"] & X' \otimes Y'
\end{tikzcd}$$

In other words, since f and g are in different components of the tensor product, it doesn't matter which order we apply them in. When arguing commutativity of a diagram, if we have something of the form of the square above (usually without the diagonal inserted), we will simply say that it is clear that the square commutes.

Now we will describe the appropriate functors between monoidal categories, called "monoidal functors". These should act like a homomorphism of monoids, preserving the multiplicative structure up to natural isomorphism, while also having functorial structure.

Definition 1.3. Let $(\mathcal{C}, \otimes, a, 1, l, r)$ and $(\mathcal{D}, \boxtimes, a', 1', l', r')$ be monoidal categories. Then a *monoidal functor* between them is a triple (F, J, φ), where $F : \mathcal{C} \to \mathcal{D}$ is a functor, $J(X,Y) : F(X) \boxtimes F(Y) \xrightarrow{\sim} F(X \otimes Y)$ is a natural isomorphism (called the *monoidal structure*), and $\varphi : 1 \xrightarrow{\sim} F(1)$ is an isomorphism such that the following hold:

1. The Monoidal Structure Axiom: For $X, Y, Z \in \mathcal{C}$, the following commutes:

$$\begin{tikzcd}
(F(X) \boxtimes F(Y)) \boxtimes F(Z) \arrow[r, "a'(F(X){,}F(Y){,}F(Z))"] \arrow[d, "J(X{,}Y)\boxtimes \mathrm{id}_{F(Z)}"] & F(X) \boxtimes (F(Y) \boxtimes F(Z)) \arrow[d, "\mathrm{id}_{F(X)} \boxtimes J_{(Y,Z)}"] \\
F(X \otimes Y) \boxtimes F(Z) \arrow[d, "J(X\otimes Y{,}Z)"] & F(X) \boxtimes F(Y \otimes Z) \arrow[d, "J(X{,}Y\otimes Z)"] \\
F((X \otimes Y) \otimes Z) \arrow[r, "F(a(X{,}Y{,}Z))"] & F(X \otimes (Y \otimes Z))
\end{tikzcd}$$

2. The Unit Coherence Axiom: For $X \in \mathcal{C}$, the following diagrams commute

$$\begin{tikzcd}
1' \boxtimes F(X) \arrow[r, "l'(F(X))"] \arrow[d, "\varphi \boxtimes \mathrm{id}_{F(X)}"] & F(X) \arrow[d, "F(l(X))^{-1}"] \\
F(1) \boxtimes F(X) \arrow[r, "J(1,X)"] & F(1 \otimes X)
\end{tikzcd}
\qquad
\begin{tikzcd}
F(X) \boxtimes 1' \arrow[r, "r'(F(X))"] \arrow[d, "\mathrm{id}_{F(X)} \boxtimes \varphi"] & F(X) \arrow[d, "F(r(X))^{-1}"] \\
F(X) \boxtimes F(1) \arrow[r, "J(X,1)"] & F(X \otimes 1)
\end{tikzcd}$$

Remark 1.4. If we have monoidal functors $(F, J, \varphi) : \mathcal{C} \to \mathcal{D}$ and $(G, K, \psi) : \mathcal{D} \to \mathcal{E}$ we have a composition monoidal functor $(G \circ F, G(J) \circ K(F), G(\varphi) \circ \psi) : \mathcal{C} \to \mathcal{E}$. Explicitly, the monoidal structure is given by

$$[G(J) \circ K(F)](X, Y) : G(F(X)) \otimes G(F(Y)) \xrightarrow{K(F(X),F(Y))} G(F(X) \otimes F(Y)) \xrightarrow{G(J(X,Y))} G(F(X \otimes Y))$$

In a similar manner, monoidal natural transformations are natural transformations which preserve the monoidal structure of monoidal functors.

Definition 1.5. Let $(\mathcal{C}, \otimes, a, 1, l, r)$ and $(\mathcal{D}, \boxtimes, a', 1', l', r')$ be monoidal categories, and let (F, J, φ) and (G, K, ψ) be monoidal functors $\mathcal{C} \to \mathcal{D}$. Then a natural transformation of monoidal functors is a natural transformation $\eta : F \Rightarrow G$ such that $\eta(1) : F(1) \to G(1)$ is an isomorphism, and for $X, Y \in \mathcal{C}$, the following diagrams commute:

$$\begin{tikzcd}
F(X) \boxtimes F(Y) \arrow[r, "J(X,Y)"] \arrow[d, "\eta(X) \boxtimes \eta(Y)"] & F(X \otimes Y) \arrow[d, "\eta(X \otimes Y)"] \\
G(X) \boxtimes G(Y) \arrow[r, "K(X,Y)"] & G(X \otimes Y)
\end{tikzcd}
\qquad
\begin{tikzcd}
& 1 \arrow[dl, "\varphi"'] \arrow[dr, "\psi"] & \\
F(1) \arrow[rr, "\eta(1)"] & & G(1)
\end{tikzcd}$$

Similarly to the fact that natural transformations give the collection of functors between two categories the structure of a category, monoidal natural transformations gives the collection of monoidal functors between monoidal categories the structure of a category.

Definition 1.6. Let $\mathrm{Aut}_{\otimes}(\mathcal{C})$ denote the category of monoidal auto-equivalences of a monoidal category $\mathcal{C}$.

Remark 1.7. Due to some basic results, as well as the Mac Lane Coherence/Strictness theorems, for our purposes, we may safely identify $X = X \otimes 1 = 1 \otimes X$ such that $l = r = \mathrm{id}$, as well as any association of products, e.g. $X \otimes Y \otimes Z = (X \otimes Y) \otimes Z = X \otimes (Y \otimes Z)$. If interested, one could insert the appropriate identities and natural isomorphisms into what we write, but for our purposes this would only serve to clutter and obscure. For more detailed discussions of this, see MacLane (1978) chapters 7 and 11, or chapter 2 of Etingof et al. (2015). In a similar vein, for a monoidal functor F, we will identify $F(1) = 1$, so that φ is trivial, as well as $J(1, X)$ and $J(X, 1)$.

1.2.1 Rigid Monoidal Categories

Rigid monoidal categories categorify something slightly weaker than the concept of a multiplicative inverse for a monoidal category. In a monoid, multiplication by a multiplicative inverse gives the identity, and vice versa (i.e. $mm^{-1} = 1$). We could consider an inverse to be something such that when we tensor an object with it's inverse, we get something isomorphic to the identity object (i.e. $X \otimes X^{-1} \cong 1$). Since monoidal categories have the additional structure of morphisms, we could make a weakening called a "dual" where we just have a morphism to the identity object instead of an isomorphism, and vice versa. Now however, since we can't isomorphically "cancel" things with it's dual, we see that there is a distinction between a left and a right dual. This leads us to the following definition:

Definition 1.8. Let $\mathcal{C}$ be a monoidal category, and let $X \in \mathcal{C}$:

1. An object $X^* \in \mathcal{C}$ is a *left dual* of X if there are morphisms $\mathrm{ev}_X : X^* \otimes X \to 1$ and $\mathrm{coev}_X : 1 \to X \otimes X^*$ such that following triangles commute:

$$\begin{tikzcd} X \arrow[r, "\mathrm{coev}_X \otimes \mathrm{id}"] \arrow[dr, "\mathrm{id}"'] & X \otimes X^* \otimes X \arrow[d, "\mathrm{id} \otimes \mathrm{ev}_X"] \\ & X \end{tikzcd} \qquad \begin{tikzcd} X^* \arrow[r, "\mathrm{id} \otimes \mathrm{coev}_X"] \arrow[dr, "\mathrm{id}"'] & X^* \otimes X \otimes X^* \arrow[d, "\mathrm{ev}_X \otimes \mathrm{id}"] \\ & X^* \end{tikzcd}$$

2. An object ${}^*X \in \mathcal{C}$ is a *right dual* of X if there are morphisms $\mathrm{ev}_X : X \otimes {}^*X \to 1$ and $\mathrm{coev}_X : 1 \to {}^*X \otimes X$ such that following triangles commute:

$$\begin{tikzcd} X \arrow[r, "\mathrm{id} \otimes \mathrm{coev}_X"] \arrow[dr, "\mathrm{id}"'] & X \otimes {}^*X \otimes X \arrow[d, "\mathrm{ev}_X \otimes \mathrm{id}"] \\ & X \end{tikzcd} \qquad \begin{tikzcd} {}^*X \arrow[r, "\mathrm{coev}_X \otimes \mathrm{id}"] \arrow[dr, "\mathrm{id}"'] & {}^*X \otimes X \otimes {}^*X \arrow[d, "\mathrm{id} \otimes \mathrm{ev}_X"] \\ & {}^*X \end{tikzcd}$$

3. $\mathcal{C}$ is called rigid if each object admits a left and right dual.

Remark 1.9. There is some ambiguity in notation of the maps, but in practice it should be clear which is used based on the placement of the star. Also note that we have ${}^*(X^*) \cong X \cong ({}^*X)^*$ and ${}^*1 = 1 = 1^*$.

Theorem 1.10. *Dual objects are unique up to a unique isomorphism (see Etingof et al. (2015) Proposition 2.10.5).*

Example 1.11. In the case that $\mathcal{C} = \mathrm{Vec}_{\Bbbk} = \mathrm{Vec}$ (the category of finite dimensional vector spaces (over $\Bbbk$)), then we have that ${}^*V = V^* = \mathrm{Hom}(V, \Bbbk)$, with evaluation as defined below:

$$\mathrm{ev}_V : V^* \otimes V \to \Bbbk; (f, v) \mapsto f(v)$$

For the coevaluation, fix a basis $\{e_1, ..., e_n\}$ of V, and let $\{e^1, ..., e^n\}$ be its corresponding dual basis (i.e. such that $e^i(e_j) = \delta_{i,j}$). Then we have coevaluation as below:

$$\mathrm{coev}_V : \Bbbk \to V \otimes V^*; 1 \mapsto 1 \sum_{i=1}^{n} e_i \otimes e^i$$

In the case above, we can also use the notion of duals to define the dual of a linear map. Indeed, if $f : V \to W$ is linear, we have a linear $f^* : W^* \to V^*$ by $[f^*(\varphi)](x) = \varphi(f(x))$. Notice that the below definition generalizes this to any rigid monoidal category.

Definition 1.12. Given a morphism $f : X \to Y$ in a rigid monoidal category $(\mathcal{C}, \otimes, 1)$:

1. Its *left dual* $f^* : Y^* \to X^*$ is defined as follows:

$$Y^* \xrightarrow{\mathrm{id} \otimes \mathrm{coev}_X} Y^* \otimes X \otimes X^* \xrightarrow{\mathrm{id} \otimes f \otimes \mathrm{id}} Y^* \otimes Y \otimes X^* \xrightarrow{\mathrm{ev}_Y \otimes \mathrm{id}} X^*$$

2. Its *right dual* ${}^*f : {}^*Y \to {}^*X$ is defined as follows:

$${}^*Y \xrightarrow{\mathrm{coev}_X \otimes \mathrm{id}} {}^*X \otimes X \otimes {}^*Y \xrightarrow{\mathrm{id} \otimes f \otimes \mathrm{id}} {}^*X \otimes Y \otimes {}^*Y \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_Y} {}^*Y$$

Remark 1.13. From here, it is easy to show that $(X \otimes Y)^* = Y^* \otimes X^*$ and $(f \circ g)^* = g^* \circ f^*$, as well as similar statements for right duals, with appropriate evaluations and coevaluations. Recall from basic category theory that for any category $\mathcal{C}$, we can form its *opposite category* $\mathcal{C}^\vee$ with the same objects, but for each arrow $f : X \to Y$ in $\mathcal{C}$, we have an arrow $f' : Y \to X$ in $\mathcal{C}^\vee$. For monoidal categories, there is another notion of an opposite category, its *monoidal opposite category* $\mathcal{C}^{\mathrm{op}}$, with the same underlying category, but the monoidal structure is given by $X \otimes^{\mathrm{op}} Y := Y \otimes X$. The above yeilds that $(-)^*$ and ${}^*(-)$ make up a monoidal equivalences $\mathcal{C}^\vee \xrightarrow{\sim} \mathcal{C}^{\mathrm{op}}$.

Similarly to how homomorphisms of monoids automatically preserve inverses when they exist, monoidal functors preserve duals. One can also say something about the dual's component of a natural transformation.

Theorem 1.14. *Let $(F, J) : \mathcal{C} \to \mathcal{D}$ be a monoidal functor between rigid monoidal categories. Then if $X \in \mathcal{C}$, $F(X)^* = F(X^*)$ with the following maps:*

$$\begin{aligned}
\mathrm{ev}_{F(X)} :& F(X^*) \otimes F(X) \xrightarrow{J(X^*,X)} F(X^* \otimes X) \xrightarrow{F(\mathrm{ev}_X)} F(1) = 1 \\
\mathrm{coev}_{F(X)} :& 1 = F(1) \xrightarrow{F(\mathrm{coev}_X)} F(X \otimes X^*) \xrightarrow{J(X,X^*)^{-1}} F(X) \otimes F(X^*)
\end{aligned}$$

A similar statement also holds for right duals.

Proof. For this, refer to the below diagram:

$$\begin{tikzcd}
F(X \otimes X^*) \otimes F(X) \arrow[rr, "J(X,X^*)^{-1} \otimes \mathrm{id}"] \arrow[drr, "J(X \otimes X^*, X)"] & & F(X) \otimes F(X^*) \otimes F(X) \arrow[rr, "\mathrm{id} \otimes J(X^*,X)"] & & F(X) \otimes F(X^* \otimes X) \arrow[d, "F(\mathrm{id}) \otimes F(\mathrm{ev}_X)"] \\
F(X) \arrow[u, "F(\mathrm{coev}_X) \otimes F(\mathrm{id})"] \arrow[rr, "F(\mathrm{coev}_X \otimes \mathrm{id})"] & & F(X \otimes X^* \otimes X) \arrow[urr, "J(X, X^* \otimes X)^{-1}"] \arrow[rr, "F(\mathrm{ev}_X \otimes \mathrm{id})"] & & F(X)
\end{tikzcd}$$

The path from the bottom left, along the top, and then down to the bottom right is

$$(\mathrm{id}_{F(X)} \otimes \mathrm{ev}_{F(X)}) \circ (\mathrm{coev}_{F(X)} \otimes \mathrm{id}_{F(X)})$$

Meanwhile, the bottom path is $F(\mathrm{id}_X) = \mathrm{id}_{F(X)}$. The middle quadrilateral commutes by the monoidal structure axiom for F, while the two triangles on the sides commutes by naturality of J (where since we identify $1 \otimes X = X$ and $F(1) \otimes F(X) = 1 \otimes F(X) = F(X)$, one side of the naturality square is identity and hence omitted). Thus one of the two necessary triangles commute for $F(X^*)$ to be a left dual of $F(X)$. The other is similar. $\square$

Corollary 1.15. With the same setup as the previous theorem, if $f : X \to Y$ is a morphism in $\mathcal{C}$, then $F(f)^* = F(f^*)$. A similar statement holds for right duals.

Proof. In the below diagram, the path from the bottom left, along the top, and down to the bottom right is $F(f)^*$, meanwhile the bottom path is $F(f^*)$. So it suffices to show it commutes.

$$\begin{tikzcd}
F(Y^*) \otimes F(X \otimes X^*) \arrow[r, "\mathrm{id} \otimes J(X,X^*)^{-1}"] \arrow[dr, "J(Y^*,X\otimes X^*)"] & F(Y^*) \otimes F(X) \otimes F(X^*) \arrow[r, "\mathrm{id} \otimes F(f) \otimes \mathrm{id}"] & F(Y^*) \otimes F(Y) \otimes F(X^*) \arrow[r, "J(Y^*,Y)\otimes \mathrm{id}"] & F(Y^* \otimes Y) \otimes F(X^*) \arrow[d, "F(\mathrm{ev}_Y)\otimes F(\mathrm{id})"] \\
F(Y^*) \arrow[u, "F(\mathrm{id})\otimes F(\mathrm{coev}_X)"] \arrow[r, "F(\mathrm{id} \otimes \mathrm{coev}_X)"] & F(Y^* \otimes X \otimes X^*) \arrow[r, "F(\mathrm{id} \otimes f \otimes \mathrm{id})"] & F(Y^* \otimes Y \otimes X^*) \arrow[r, "F(\mathrm{ev}_Y \otimes \mathrm{id})"] \arrow[ur, "J(Y^*\otimes Y,X^*)^{-1}"] & F(X^*)
\end{tikzcd}$$

The left and right triangles commute again like in the proof of Theorem 1.14. For the middle, notice that it is equivalent (after taking some inverses) to the perimiter of the following diagram:

$$\begin{tikzcd}
 & F(Y^*) \otimes F(X) \otimes F(X^*) \arrow[r, "\mathrm{id} \otimes F(f) \otimes \mathrm{id}"] \arrow[dl, "\mathrm{id} \otimes J(X,X^*)"'] \arrow[d, "J(Y^*,X)\otimes \mathrm{id}"] & F(Y^*) \otimes F(Y) \otimes F(X^*) \arrow[d, "J(Y^*,Y)\otimes \mathrm{id}"] \\
F(Y^*) \otimes F(X \otimes X^*) \arrow[dr, "J(Y^*,X\otimes X^*)"'] & F(Y^* \otimes X) \otimes F(X^*) \arrow[r, "F(\mathrm{id} \otimes f)\otimes \mathrm{id}"] \arrow[d, "J(Y^*\otimes X,Y^*)"] & F(Y^* \otimes Y) \otimes F(X^*) \arrow[d, "J(Y^*\otimes Y,X^*)"] \\
 & F(Y^* \otimes X \otimes X^*) \arrow[r, "F(\mathrm{id} \otimes f \otimes \mathrm{id})"] & F(Y^* \otimes Y \otimes X^*)
\end{tikzcd}$$

The right two squares commute via naturality of J, and the left quadrilateral commutes by the monoidal structure axiom. Hence the middle part of the first diagram commutes, as desired. $\square$

Corollary 1.16. Let $(F, J), (G, K) : \mathcal{C} \to \mathcal{D}$ be monoidal functors between rigid categories, and let $\eta : F \Rightarrow G$ be a natural isomorphism of monoidal functors. Then $(\eta(X)^{-1})^* = \eta(X^*)$.

Proof. We will show that $\eta(X)^* = \eta(X^*)^{-1}$, which is equivalent by applying that to η^{-1}. Let $X, Y, Z \in \mathcal{C}$, and refer to the below diagram:

$$\begin{tikzcd}
G(X) \otimes F(Y) \otimes F(Z) \arrow[rrrr, "\mathrm{id} \otimes \eta(Y) \otimes \mathrm{id}"] \arrow[drr, "\mathrm{id} \otimes \eta(Y) \otimes \eta(Z)"'] & & & & G(X) \otimes G(Y) \otimes F(Z) \arrow[dll, "\mathrm{id} \otimes \mathrm{id} \otimes \eta(Z)"] \arrow[ddd, "K(X,Y)\otimes \mathrm{id}"'] \\
 & & G(X) \otimes G(Y) \otimes G(Z) \arrow[dl, "\mathrm{id} \otimes K(Y,Z)"] \arrow[dr, "K(X,Y)\otimes \mathrm{id}"'] & & \\
 & G(X) \otimes G(Y \otimes Z) \arrow[dr, "K(X,Y\otimes Z)"] & & G(X \otimes Y) \otimes G(Z) \arrow[dl, "K(X\otimes Y,Z)"'] \arrow[dr, "\mathrm{id} \otimes \eta(Z)^{-1}"] & \\
G(X) \otimes F(Y \otimes Z) \arrow[uuu, "\mathrm{id} \otimes J(X,Y)^{-1}"'] \arrow[ur, "\mathrm{id} \otimes \eta(Y\otimes Z)"] & & G(X \otimes Y \otimes Z) & & G(X \otimes Y) \otimes F(Z)
\end{tikzcd}$$

The left quadrilateral commutes since η is a monoidal natural transformation. The top triangle clearly commutes. The middle square commutes via the monoidal structure axiom for G, and finally the right quadrilateral clearly commutes.

Now, for $X \in \mathcal{C}$, see the below diagram:

$$\begin{tikzcd}
G(X^* \otimes X \otimes X^*) \arrow[rr, "K(X^* \otimes X, X^*)^{-1}"] & & G(X^* \otimes X) \otimes G(X^*) \arrow[d, "\mathrm{id} \otimes \eta(X^*)^{-1}"] \arrow[dl, "G(\mathrm{ev}_X) \otimes \mathrm{id}"'] \\
G(X^*) \otimes G(X \otimes X^*) \arrow[u, "K(X^*, X \otimes X^*)"] & & G(X^* \otimes X) \otimes F(X^*) \arrow[d, "G(\mathrm{ev}_X) \otimes \mathrm{id}"] \\
G(X^*) \otimes F(X \otimes X^*) \arrow[u, "\mathrm{id} \otimes \eta(X \otimes X^*)"] & G(X^*) \arrow[l, "\mathrm{id} \otimes F(\mathrm{coev}_X)"] \arrow[ul, "\mathrm{id} \otimes G(\mathrm{coev}_X)"] \arrow[uul, bend right, "G(\mathrm{id} \otimes \mathrm{coev}_X)"'] \arrow[r, "\eta(X^*)^{-1}"] & F(X^*)
\end{tikzcd}$$

The bottom left triangle commutes via naturality of η and the upper left triangle commutes by the naturality of K (both ommitting the trivial 1 side of the naturality squares). The right quadrilateral clearly commutes (the evaluation and η are in different tensor components but the 1 is omitted). The center triangle commutes by the definition of duals and naturality of K. The perimiter path from $G(X^*)$ to $F(X^*)$ started by going to the left is the same as $\eta(X)^*$ by using the first diagram with X^*, X, and X^*. Thus $\eta(X)^* = \eta(X^*)^{-1}$. $\square$

We now record how duals interact with hom sets. These adjunctions correspond to how one can "multiply by the inverse on both sides", e.g. $xy = z \iff y = x^{-1}z$.

Theorem 1.17. *Let $\mathcal{C}$ be a rigid monoidal category with $V \in \mathcal{C}$. There are adjunctions:*

1. $\mathrm{Hom}_{\mathcal{C}}(U \otimes V, W) \cong \mathrm{Hom}_{\mathcal{C}}(U, W \otimes V^*)$
2. $\mathrm{Hom}_{\mathcal{C}}(V^* \otimes U, W) \cong \mathrm{Hom}_{\mathcal{C}}(U, V \otimes W)$
3. $\mathrm{Hom}_{\mathcal{C}}(U \otimes {}^*V, W) \cong \mathrm{Hom}_{\mathcal{C}}(U, W \otimes V)$
4. $\mathrm{Hom}_{\mathcal{C}}({}^*V \otimes U, W) \cong \mathrm{Hom}_{\mathcal{C}}(U, V \otimes W)$

Proof. We will only prove part 1 since the others are similar. Let $f : U \otimes V \to W$. Define $\phi(f)$ as the composition below:

$$U \xrightarrow{\mathrm{id} \otimes \mathrm{coev}_V} U \otimes V \otimes V^* \xrightarrow{f \otimes \mathrm{id}} W \otimes V^*$$

Then ϕ defines a function $\mathrm{Hom}_{\mathcal{C}}(U \otimes V, W) \to \mathrm{Hom}_{\mathcal{C}}(U, W \otimes V^*)$. Similarly for $g : U \to W \otimes V^*$ define $\psi(g)$ as the following:

$$U \otimes V \xrightarrow{g \otimes \mathrm{id}} W \otimes V^* \otimes V \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_V} W$$

Then ψ defines a function $\mathrm{Hom}_{\mathcal{C}}(U, W \otimes V^*) \to \mathrm{Hom}_{\mathcal{C}}(U \otimes V, W)$. Consider the following diagram:

$$\begin{tikzcd}
U \otimes V \arrow[r, "\phi(f) \otimes \mathrm{id}"] \arrow[d, "\mathrm{id} \otimes \mathrm{coev}_V \otimes \mathrm{id}"'] & W \otimes V^* \otimes V \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_V"] & W \\
U \otimes V \otimes V^* \otimes V \arrow[ur, "f \otimes \mathrm{id} \otimes \mathrm{id}"] \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{ev}_V"] & U \otimes V \arrow[ur, "f"] &
\end{tikzcd}$$

Notice that the triangle on the left commutes by definition of $\phi(f)$ and the quadrilateral on the right clearly commutes. By the definition of left duals, the path on the bottom is f, meanwhile the top path is $\psi(\phi(f))$. Thus $\psi \circ \phi = \mathrm{id}$. The other composition is similar. Thus ϕ and ψ are mutually inverse isomorphisms. It is left to show naturality.

Let $h : U' \to U$, $k : W \to W'$, and $f : U \otimes V \to W$. Then we have the diagram below:

$$\begin{tikzcd}
U' \arrow[r, "h"] \arrow[dr, "\mathrm{id} \otimes \mathrm{coev}_V"'] & U \arrow[r, "\mathrm{id} \otimes \mathrm{coev}_V"] & U \otimes V \otimes V^* \arrow[r, "f \otimes \mathrm{id}"] & W \otimes V^* \arrow[r, "k \otimes \mathrm{id}"] & W \otimes V^* \\
& U' \otimes V \otimes V^* \arrow[ur, "h \otimes \mathrm{id}"] \arrow[urrr, "(h \circ f \circ k) \otimes \mathrm{id}"'] & & &
\end{tikzcd}$$

This clearly commutes, and taking the outer perimeter yeilds that

$(\mathrm{Hom}(h, k \otimes \mathrm{id}) \circ \phi)(f) = (\phi \circ \mathrm{Hom}(k, h))(f)$. Thus the isomorphism is natural. □

Remark 1.18. Note that part 1 implies that rigid monoidal categories have an internal hom (see chapter 3) with respect to their product, since the functor $- \otimes V$ has a right adjoint $- \otimes V^*$.

Example 1.19. In the situation of Example 1.11, we have that $W \otimes V^* \cong \mathrm{Hom}(V, W)$ by $w \otimes \alpha \mapsto \alpha(-)w$. Then translating part 1 of the above theorem we get an adjunction:

$$\mathrm{Hom}_{\mathrm{Vec}}(U \otimes V, W) \cong \mathrm{Hom}_{\mathrm{Vec}}(U, \mathrm{Hom}_{\mathrm{Vec}}(V, W))$$

Where $\phi(f)(u)(v) = f(u \otimes v)$ and $\psi(g)(u \otimes v) = g(u)(v)$.

1.3 Tensor Categories

Just how monoidal categories generalized monoids "up to natural isomorphism", tensor categories generalize rings. We will have a monoidal structure, but also an additive structure, in the form of a direct sum. Unlike monoidal categories though, we will have some homological structure, i.e. images, kernals, and cokernals. This follows the fashion of abelian categories, which seek not only to emulate the behavior of an abelian group, but also the category of abelian groups. Tensor categories similarly don't just generalize rings, but due to their abelian category structure, also generalize the category of vector spaces, and the categories of representations of a finite group or a Hopf algebra.

We will be extremely brief on our introduction to abelian, tensor, and fusion categories. For greater detail and exploration, see Kashiwara and Schapira (2005) and Etingof et al. (2015).

Definition 1.20. An *additive category* is a category $\mathcal{C}$ such that:

1. For each $X, Y \in \mathcal{C}$, $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is an abelian group such that composition is bi-additive.
2. There is an object $0 \in \mathcal{C}$ such that $\mathrm{Hom}_{C}(0, 0)$ is the trivial group.
3. For any $X, Y \in \mathcal{C}$, the biproduct $X \oplus Y$ exists, this is called the direct sum of X and Y.

Definition 1.21. A $\Bbbk$-*linear category* is an additive category $\mathcal{C}$ such that $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is moreover a vector space over $\Bbbk$, and composition is $\Bbbk$-linear.

Definition 1.22. A functor $F : \mathcal{C} \to \mathcal{D}$ between additive (resp. $\Bbbk$-linear) categories is *additive* (resp. $\Bbbk$-*linear*) if for each $X, Y \in \mathcal{C}$, the following induced map is a homomorphism of abelian groups (resp. $\Bbbk$-linear):

$$\mathrm{Hom}_{\mathcal{C}}(X, Y) \to \mathrm{Hom}_{\mathcal{D}}(F(X), F(Y))$$

Remark 1.23. Any additive functor induces a natural isomorphism $F(\bigoplus_i X_i) \cong \bigoplus_i F(X_i)$ for a finite sum, i.e. they naturally preserve finite direct sums.

Definition 1.24. Let $\mathcal{C}$ be an additive category, and $f : X \to Y$ a morphism in $\mathcal{C}$.

1. If it exists, the *kernel* of f, $\mathrm{Ker}(f)$, is the equalizer of $f, 0 : X \rightrightarrows Y$.

2. If it exists, the *cokernel* of f, $\mathrm{Coker}(f)$, is the coequalizer of $f, 0 : X \rightrightarrows Y$.

Definition 1.25. An *abelian category* is an additive category $\mathcal{C}$ such that every morphism in $\mathcal{C}$, $\varphi : X \to Y$, $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ exist, and furthermore that φ decomposes as $X \xrightarrow{i} \mathrm{Im}(\varphi) \xrightarrow{j} Y$ such that $\mathrm{Coker}(k) = \mathrm{Im}(\varphi) = \mathrm{Ker}(c)$ where k and c are the morphisms associated to $\mathrm{Ker}(\varphi)$ and $\mathrm{Coker}(\varphi)$ respectively.

Remark 1.26. This is essentially the categorical structure one needs to define exact sequences.

Let us get some basic definitions regarding abelian categories sorted out.

Definition 1.27. Let $\{\mathcal{C}_\alpha\}_{\alpha \in I}$ be an I-indexed collection of abelian (or $\mathbb{k}$-linear abelian or additive) categories. Then we can form an abelian ($\mathbb{k}$-linear abelian/additive) category called the *direct sum* $\bigoplus_{\alpha \in I} \mathcal{C}_\alpha$ of $\{\mathcal{C}_\alpha\}_{\alpha \in I}$. The objects are formal sums $\bigoplus_{\alpha \in I} X_\alpha$ with $X_\alpha \in \mathcal{C}_\alpha$ and all but finitely many are zero. The morphisms are $\mathrm{Hom}_{\bigoplus_{\alpha \in I} \mathcal{C}_\alpha}(X, Y) := \bigoplus_{\alpha \in I} \mathrm{Hom}_{\mathcal{C}_\alpha}(X_\alpha, Y_\alpha)$.

Definition 1.28. Let $\mathcal{C}$ be abelian.

1. A morphism $f : X \to Y$ is a *monomorphism* if $\mathrm{Ker}(f) = 0$, and an *epimorpihsm* if $\mathrm{Coker}(f) = 0$.

2. A *subobject*, X, of $Y \in \mathcal{C}$ is a monomorphism $i : X \to Y$. We write $X \subseteq Y$. A quotient, Z, of Y is an epimorpihsm $p : Y \to Z$. If $i : X \subseteq Y$, then we have a quotient Y/X of Y given by $\mathrm{Coker}(i)$.

3. A nonzero $X \in \mathcal{C}$ is *simple* if 0 and X are its only subobjects, and *semisimple* if it is a direct sum of simple objects. $\mathcal{C}$ is called *semisimple* if every object is semisimple.

4. We say that X has *finite length* if there exists a sequence of subobjects as below such that X_i/X_{i-1} is simple for each i:

$$0 = X_0 \subseteq X_1 \subseteq ... \subseteq X_{n-1} \subseteq X_n = X$$

5. We say that $P \in \mathcal{C}$ is *projective* if the functor $\mathrm{Hom}_{\mathcal{C}}(P, -)$ is exact. A *projective cover* of X is a projective P and an epimorpihsm $p : P \to X$ such that for any other epimorphism $p' : P' \to X$ from a projective object, there is an epimorphism $h : P' \to P$ such that $p \circ h = p'$.

Definition 1.29. Let $\mathcal{C}$ be $\Bbbk$-linear abelian.

1. We call $\mathcal{C}$ *locally finite* if:

   (a) For all $X, Y \in \mathcal{C}$, $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ is finite dimensional

   (b) Every object $X \in \mathcal{C}$ has finite length

2. We call $\mathcal{C}$ *finite* if:

   (a) $\mathcal{C}$ is locally finite

   (b) Every simple object of $\mathcal{C}$ has a projective cover

   (c) There are finitely many isomorphism classes of simple objects

Remark 1.30. Finiteness is the same as being equivalent to the category of finite dimensional modules over a finite dimensional $\Bbbk$-algebra.

Definition 1.31. A *tensor category* is a locally finite $\Bbbk$-linear rigid monoidal category $\mathcal{C}$ such that $\otimes : \mathcal{C} \times \mathcal{C} \to \mathcal{C}$ is bilinear on morphisms and $\mathrm{Hom}_{\mathcal{C}}(1, 1) \cong \Bbbk$. A *fusion category* is a finite semisimple tensor category.

Remark 1.32. In fact, in the above definitions, $\otimes$ will be biexact.

Definition 1.33. Let $\mathcal{C}, \mathcal{D}$ be tensor categories. A *tensor functor* $F : \mathcal{C} \to \mathcal{D}$ is an exact faithful $\Bbbk$-linear monoidal functor (F, J) with $F(1) = 1$. Let $\mathrm{Aut}_{\otimes}(\mathcal{C})$ be the monoidal category of tensor auto-equivalences of $\mathcal{C}$ and monoidal natural transformations between them.

Definition 1.34. A *tensor subcategory* of a tensor category $\mathcal{C}$ is a full subcategory which is closed under tensor products, duals, and quotients of subobjects (called *subquotients*).

1.4 Module Categories

Retaining the theme of categorification, we now introduce the notion of a module category over a monoidal category. This generalizes the notion of a monoid action on a set. We again only hold things up to natural isomorphism, asking that the action respects the multiplication and multiplicative identity of a monoidal category up to a natural isomorphism, with some coherency conditions. We present the definitions for left modules, but right modules are similar.

Definition 1.35. Let $\mathcal{C}$ be a monoidal category. A *(left) module category over* $\mathcal{C}$ is a quadruple $(\mathcal{M}, \otimes^{\mathcal{M}}, m, l)$, where $\mathcal{M}$ is a category with an action bifunctor $\otimes^{\mathcal{M}} : \mathcal{C} \times \mathcal{M} \to \mathcal{M}$, a natural module associativity constraint isomorphism $m(X,Y,Z) : (X \otimes Y) \otimes^{\mathcal{M}} M \xrightarrow{\sim} X \otimes^{\mathcal{M}} (Y \otimes^{\mathcal{M}} M)$, and a natural unit constraint isomorphism $l(M) : 1 \otimes^{\mathcal{M}} M \xrightarrow{\sim} M$ such that the following hold:

1. The (Module) Pentagon Axiom: For $X, Y, Z \in \mathcal{C}$ and $Z \in \mathcal{M}$, the following commutes:

$$\begin{tikzcd}
 & ((X \otimes Y) \otimes Z) \otimes^{\mathcal{M}} M \arrow[dl, "a(X,Y,Z)\otimes \mathrm{id}"'] \arrow[dr, "m(X\otimes Y,Z,M)"] & \\
(X \otimes (Y \otimes Z)) \otimes^{\mathcal{M}} M \arrow[d, "m(X,Y\otimes Z,M)"] & & (X \otimes Y) \otimes^{\mathcal{M}} (Z \otimes^{\mathcal{M}} M) \arrow[d, "m(X,Y,Z\otimes^{\mathcal{M}} M)"] \\
X \otimes^{\mathcal{M}} ((Y \otimes Z) \otimes^{\mathcal{M}} M) \arrow[rr, "\mathrm{id} \otimes m_{Y,Z,M}"] & & X \otimes^{\mathcal{M}} (Y \otimes^{\mathcal{M}} (Z \otimes^{\mathcal{M}} M))
\end{tikzcd}$$

2. The (Module) Triangle Axiom: For $X \in \mathcal{C}$, $M \in \mathcal{M}$, the following commutes:

$$\begin{tikzcd}
(X \otimes 1) \otimes^{\mathcal{M}} M \arrow[rr, "m(X,1,M)"] \arrow[dr, "r(X)\otimes \mathrm{id}"] & & X \otimes^{\mathcal{M}} (1 \otimes^{\mathcal{M}} M) \arrow[dl, "\mathrm{id} \otimes l(M)"] \\
 & X \otimes^{\mathcal{M}} M &
\end{tikzcd}$$

Remark 1.36. We call $\otimes^{\mathcal{M}}$ the *action of* $\mathcal{C}$ *on* $\mathcal{M}$, denoting the tensor product with a superscript to distinguish it from the tensor product in $\mathcal{C}$. Note that we don't do so for tensoring morphisms. In practice, the author has found that putting the superscript on the tensor product in commutative diagrams offers more clutter than clarity.

We now categorify the notion of a homomorphism of M-sets (or sets equipped with an action of M) for a monoid M. These are functorial, and preserve the action up to natural isomorphism.

Definition 1.37. Let $\mathcal{C}$ a monoidal category, and let $(\mathcal{M}, \otimes^{\mathcal{M}}, m, l), (\mathcal{N}, \otimes^{\mathcal{N}}, n, l')$ be (left) module categories over $\mathcal{C}$. A *$\mathcal{C}$-module functor* is a functor $F : \mathcal{M} \to \mathcal{N}$ and a natural isomorphism, called the module structure , $s(X, M) : F(X \otimes^{\mathcal{M}} M) \xrightarrow{\sim} X \otimes^{\mathcal{N}} F(M)$ such that for $X, Y \in \mathcal{C}$ and $M \in \mathcal{M}$, the following diagrams commute:

$$\begin{tikzcd}
 & F((X \otimes Y) \otimes^{\mathcal{M}} M) \arrow[dl, "F(m(X,Y,M))"'] \arrow[dr, "s(X\otimes Y,M)"] & \\
F(X \otimes^{\mathcal{M}} (Y \otimes^{\mathcal{M}} M)) \arrow[d, "s(X,Y\otimes M)"] & & (X \otimes Y) \otimes^{\mathcal{N}} F(M) \arrow[d, "n(X,Y,F(M))"] \\
X \otimes^{\mathcal{N}} F(Y \otimes^{\mathcal{M}} M) \arrow[rr, "\mathrm{id} \otimes s(Y,M)"] & & X \otimes^{\mathcal{N}} (Y \otimes^{\mathcal{N}} F(M))
\end{tikzcd}$$

$$\begin{tikzcd}
F(1 \otimes^{\mathcal{M}} M) \arrow[rr, "s(1,M)"] \arrow[dr, "F(l(M))"] & & 1 \otimes^{\mathcal{N}} F(M) \arrow[dl, "l'(F(M))"] \\
 & F(M) &
\end{tikzcd}$$

Remark 1.38. If we have $\mathcal{C}$-module functors $(F, s) : \mathcal{M} \to \mathcal{N}$ and $(G, t) : \mathcal{N} \to \mathcal{L}$ we can compose to get a $\mathcal{C}$-module functor $(G \circ F, t \circ G(s)) : \mathcal{M} \to \mathcal{L}$. Explicitly, we have

$$[t \circ G(s)](X, M) : G(F(X \otimes^{\mathcal{M}} M)) \xrightarrow{G(s(X,M))} G(X \otimes^{\mathcal{N}} F(M)) \xrightarrow{t(X,F(M))} X \otimes^{\mathcal{L}} G(F(M))$$

Definition 1.39. Let $\mathcal{C}$ be a monoidal category, $\mathcal{M}, \mathcal{N}$ be $\mathcal{C}$-module categories, and $(F, s), (G, t)$ be $\mathcal{C}$-module functors $\mathcal{M} \to \mathcal{N}$. Then a *$\mathcal{C}$-module natural transformation* between these is a natural transformation $\eta : F \Rightarrow G$ such that for $X \in \mathcal{C}$, $M \in \mathcal{M}$, the following commutes:

$$\begin{tikzcd}
F(X \otimes^{\mathcal{M}} M) \arrow[r, "s(X,M)"] \arrow[d, "\eta(X\otimes M)"] & X \otimes^{\mathcal{N}} F(M) \arrow[d, "\mathrm{id} \otimes \eta(M)"] \\
G(X \otimes^{\mathcal{M}} M) \arrow[r, "t(X,M)"] & X \otimes^{\mathcal{N}} G(M)
\end{tikzcd}$$

Remark 1.40. Similar to Remark 1.7, developing the theory allows us to identify $1 \otimes^{\mathcal{M}} M$ with M, assuming $l = \mathrm{id}_M$. Using our identifications, the module triangle axiom asserts that $m(X, 1, M) = \mathrm{id}_{X \otimes^{\mathcal{M}} M}$ (one can derive also that $m(1, X, M) = \mathrm{id}_{X \otimes^{\mathcal{M}} M}$), and the triangle diagram for module functors asserts that $s(1, M) = \mathrm{id}_{F(M)}$.

Similarly to how if we have inverses in a monoid acting on a set, we can “act on both sides” to cancel (e.g. $m^{-1}.x = y \iff x = m.y$), module categories interact with actions by a dual:

Theorem 1.41. *Let $\mathcal{C}$ be a rigid monoidal category, $\mathcal{M}$ a (left) $\mathcal{C}$-module category, and with $X \in \mathcal{C}$. There are adjunctions:*

1. $\mathrm{Hom}_{\mathcal{M}}(X^* \otimes^{\mathcal{M}} M, N) \cong \mathrm{Hom}_{\mathcal{M}}(M, X \otimes^{\mathcal{M}} N)$
2. $\mathrm{Hom}_{\mathcal{M}}({}^*X \otimes^{\mathcal{M}} M, N) \cong \mathrm{Hom}_{\mathcal{M}}(M, X \otimes^{\mathcal{M}} N)$

Proof. The proof is essentially the same as Theorem 1.17. □

Now analogously to upgrading the definition of an action of a monoid on a set to a module over a ring, we define module categories over a tensor category. In order to do this, we require the action to respect the additive structure in the form of biexactness.

Definition 1.42. Let $\mathcal{C}$ be a tensor category over $\Bbbk$. A *module category $\mathcal{M}$ over $\mathcal{C}$* is a locally finite $\Bbbk$-linear abelian category $\mathcal{M}$ which is a (monoidal) $\mathcal{C}$-module category such that the functor $\otimes^{\mathcal{M}} : \mathcal{C} \times \mathcal{M} \to \mathcal{M}$ is biexact. We require module functors to be $\Bbbk$-linear to respect the additive structure.

CHAPTER 2. EQUIVARIANCE

Equivariance is a general notion that means the preservation of an action. If G is a group, and $(X,\rho),(Y,\pi)$ are G-sets, then an equivariant map $f : X \to Y$ is the same as a homomorphism of G-sets: for all $g \in G$, $f \circ \rho(g) = \pi(g) \circ f$, i.e. the following diagram commutes for each $g \in G$:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "\rho(g)"] & Y \arrow[d, "\pi(g)"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

In topology, equivariant maps are the appropriate notion of a homomorphism of G-spaces. In representation theory, if instead we take $(V,\rho),(W,\pi)$ to be representations of G, then the above recovers the familiar definition of an intertwining operator, or a homomorphism of representations.

2.1 Equivariantization of a Category

We wish to generalize the situations of G-sets, and G-representations to taking value in a monoidal category, or a tensor category respectively. In fact, we further generalize so that the maps $\rho(g)$ may not be automorphisms of X, they will instead be isomorphisms $T_g(X) \xrightarrow{\sim} X$ so that the domain depends on G. In order to effectively compose these, we then need that T_g is a functor. Also, in order to generalize the fact that ρ is an action of G, we need some way to relate T_{gh} and $T_g \circ T_g$. For this, we use the following notion of a group action on a monoidal/tensor category.

Definition 2.1. Let G be a finite group. We then define $\mathrm{Cat}(G)$ to be the monoidal category whose objects are the elements of G, and whose only morphisms are the identities. Then an action of G on a monoidal category $\mathcal{C}$ is a monoidal functor:

$$(T,\gamma) : \mathrm{Cat}(G) \to \mathrm{Aut}_{\otimes}(\mathcal{C})$$

If $\mathcal{C}$ is a tensor category, we have a similar definition as above, but now $\mathrm{Aut}_{\otimes}(\mathcal{C})$ denotes tensor autoequivalences.

Let G act on a monoidal/tensor category $\mathcal{C}$ by (T, γ) as above. For $g \in G$, let $T_g : \mathcal{C} \to \mathcal{C}$ denote the functor underlying $T(g) = (T_g, t_g)$, and write $\gamma_{g,h} : T_g \circ T_h \cong T_{gh}$ for the monoidal functor structure of T. Now we define equivariant objects, equivariant maps, and equivariantization.

Definition 2.2. A *G-equivarient object in $\mathcal{C}$* is a pair (X, u) where $X \in \mathcal{C}$, and $u = \{u_g : T_g(X) \xrightarrow{\sim} X\}_{g \in G}$ is a set of isomorphisms in $\mathcal{C}$ such that the following commutes for all $g, h \in G$:

$$\begin{tikzcd} T_g(T_h(X)) \arrow[r, "T_g(u_h)"] \arrow[d, "\gamma_{g,h}(X)"] & T_g(X) \arrow[d, "u_g"] \\ T_{gh}(X) \arrow[r, "u_{gh}"] & X \end{tikzcd}$$

We define $\mathcal{C}^G$ to be the category of G-equivarient objects, this is called the *equivariantization of $\mathcal{C}$*. A morphism $f : (X, u) \to (Y, v)$ in $\mathcal{C}^G$, called a *G-equivariant map*, is a morphism $f : X \to Y$ in $\mathcal{C}$ such that the following commutes for all $g \in G$:

$$\begin{tikzcd} T_g(X) \arrow[r, "T_g(f)"] \arrow[d, "u_g"] & T_g(Y) \arrow[d, "v_g"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

Here, $\mathcal{C}^G$ will also be a monoidal category with product given as below:

$$(X, u) \otimes (Y, v) := (X \otimes Y, u \otimes v)$$

$$[u \otimes v]_g : T_g(X \otimes Y) \xrightarrow{t_g^{-1}(X,Y)} T_g(X) \otimes T_g(Y) \xrightarrow{u_g \otimes u_h} X \otimes Y$$

Theorem 2.3. *If $\mathcal{C}$ is a fusion category, then so is $\mathcal{C}^G$ (see Drinfeld et al. (2010) Theorem 4.18 and Etingof et al. (2015) Section 4.15). Needs $\Bbbk$ algebraically closed and* $\mathrm{char}(\Bbbk) = 0$.

Remark 2.4. Let us see what the dual objects in $\mathcal{C}^G$ are. We claim that $(X, u)^* = (X^*, (u^{-1})^*)$, although we will only show that this indeed defines an equivariant structure. Note that for each

$g \in G$, by Theorem 1.14 we know that $T_g(X)^* = T_g(X^*)$. Hence since $u_g^{-1} : X \to T_g(X)$, we have that $(u_g^{-1})^* : T_g(X^*) \to X^*$. Thus by taking the inverse of the diagram for equivariance for (X, u), then taking the dual and using Theorem 1.14 and Corollary 1.15, we have the following commuting diagram for $g, h \in G$:

$$\begin{tikzcd} T_g(T_h(X^*)) \arrow[r, "T_g((u_h^{-1})^*)"] \arrow[d, "(\gamma_{g,h}(X)^{-1})^*"] & T_g(X^*) \arrow[d, "(u_g^{-1})^*"] \\ T_{gh}(X^*) \arrow[r, "(u_{gh}^{-1})^*"] & X^* \end{tikzcd}$$

And notice that $(\gamma_{g,h}(X)^{-1})^* = \gamma_{g,h}(X^*)$ by Corollary 1.16. Thus $(u_g^{-1})^*$ is an equivariant structure on X^*.

Corollary 2.5. It immediately follows from the above remark and Theorem 1.17 that we have an adjunction:

$$\mathrm{Hom}_{\mathcal{C}^G}((X \otimes Y, u \otimes v), (Z, w)) \cong \mathrm{Hom}_{\mathcal{C}^G}((X, u), (Z \otimes Y^*, w \otimes (v^{-1})^*))$$

In other words, $(Z \otimes Y^*, w \otimes (v^{-1})^*)$ is the internal hom from (Y, v) to (Z, w) in $\mathcal{C}^G$ (see chapter 3).

Example 2.6. Consider the case where $\mathcal{C} = \mathrm{Vec}$. Then we have that $\mathrm{Aut}_\otimes(\mathrm{Vec}) = \{\mathrm{Id}_{\mathrm{Vec}}\}$, so we have that the only action of a finite group G is trivial, i.e. $T_g(X) = X$ for $g \in G$, $X \in \mathrm{Vec}$ and $\gamma_{g,h} = \mathrm{Id}_{\mathrm{Id}_{\mathrm{Vec}}}$. Then an object of Vec^G is a vector space V with maps $\rho_g : V \xrightarrow{\sim} V$ for each $g \in G$ such that $\rho_g \circ \rho_h = \rho_{gh}$. In other words, a *representation of* G. If $(V, \rho), (W, \theta)$ are two such representations, then a G-equivarient map is a linear map $f : V \to W$ such that $f \circ \rho_g = \theta_g \circ f$ for each $g \in G$. I.e., it's an intertwining operator, or a morphism of G-representations. Thus $\mathrm{Vec}^G = \mathrm{Rep}(G)$ is just the category of finite dimensional representations of G.

2.2 Equivariant Module Categories

Now we wish to explore module categories over an equivariantization $\mathcal{C}^G$ of a fusion category. We follow Etingof et al. (2011), however a few definitions needed to be changed. The key insight is that we can define structure on a $\mathcal{C}$-module category which allows us to define equivariant objects and equivariant maps. From there, we can form the equivariantization of the module category, upgrading it to a $\mathcal{C}^G$-module category. Firstly, we define a way for our group action T to interact with the module category:

Let $\mathcal{C}$ be a fusion category, and $\mathcal{M}$ be a $\mathcal{C}$-module category with action:

$$\otimes^{\mathcal{M}} : \mathcal{C} \times \mathcal{M} \to \mathcal{M}$$

and associativity constraint:

$$m(X, Y, M) : (X \otimes Y) \otimes^{\mathcal{M}} M \xrightarrow{\sim} X \otimes^{\mathcal{M}} (Y \otimes^{\mathcal{M}} M)$$

Given $a \in \mathrm{Aut}_{\otimes}(\mathcal{C})$ with monoidal structure $b_{X,Y} : a(X) \otimes a(Y) \xrightarrow{\sim} a(X \otimes Y)$, we get a new $\mathcal{C}$-module category $\mathcal{M}^a$, whose underlying category is $\mathcal{M}$ by twisting the action of $\mathcal{M}$ by a:

$$\otimes^a : \mathcal{C} \times \mathcal{M}^a \to \mathcal{M}^a; X \otimes^a M := a(X) \otimes^{\mathcal{M}} M$$

This has the associativity constraint:

$$\begin{aligned} m^a(X, Y, M) : (X \otimes Y) \otimes^a M = a(X \otimes Y) \otimes^{\mathcal{M}} M &\xrightarrow{b(X,Y)^{-1} \otimes \mathrm{id}} [a(X) \otimes a(Y)] \otimes^{\mathcal{M}} M \\ &\xrightarrow{m(a(X), a(Y), M)} a(X) \otimes^{\mathcal{M}} (a(Y) \otimes^{\mathcal{M}} M) = X \otimes^a (Y \otimes^a M) \end{aligned}$$

Given a $\mathcal{C}$-module functor $(F, s) : \mathcal{M} \to \mathcal{N}$, we can also form a twisted module functor:

$$F^a = (F, s^a) : \mathcal{M}^a \to \mathcal{N}^a; s^a(X, M) := s(a(X), M)$$

Notice that for composable functors we have that $[G \circ F]^a = G^a \circ F^a$. Now similar to twisting functors, if we have a $\mathcal{C}$-module natural transformation $\eta : (F, s) \to (G, t)$ of $\mathcal{C}$-module functors:

$$\eta^a : F^a = (F, s^a) \to (G, t^a) = G^a; \eta^a(M) = \eta(M)$$

Given a G-action (T, γ) on $\mathcal{C}$, denote $\mathcal{M}^{T_g}$ by $\mathcal{M}^g$, $\otimes^{T_g}$ by $\otimes^g$, and so on and so forth. Then γ yields $\mathcal{C}$-module equivalences for any $g, h \in G$:

$$\Gamma_{g,h} := (\mathrm{Id}_{\mathcal{M}}, \Gamma_{g,h}) : \mathcal{M}^{gh} \xrightarrow{\sim} (\mathcal{M}^g)^h$$

Where $\Gamma_{g,h}(X,M) : X \otimes^{gh} M = T_{gh}(X) \otimes^{\mathcal{M}} M \xrightarrow{\gamma_{g,h}(X)^{-1}\otimes \mathrm{id}} T_g(T_h(X)) \otimes^{\mathcal{M}} M = X \otimes^{g^h} M$

Now we can define structure on $\mathcal{M}$ that will allow us to talk about equivariant objects and equivariant maps in a way that appropriately interacts with the group action T on $\mathcal{C}$. The point of doing this is that when we go to define the equivariantization of $\mathcal{M}$ and its module structure, we can use the fact that U_g goes to the twisted module category, as defined below, in order to get a $T_g(X)$ and make use of the equivariant structure of the objects of $\mathcal{C}^G$.

Definition 2.7. A *G-equivariant $\mathcal{C}$-module category* is a triple $(\mathcal{M}, U, \mu)$, where $\mathcal{M}$ is a $\mathcal{C}$-module category, $U = \{(U_g, \tau_g) : \mathcal{M} \xrightarrow{\sim} \mathcal{M}^g\}_{g\in G}$ is a set of $\mathcal{C}$-module equivalences, and $\mu = \{\mu_{g,h} : (U_g)^h \circ U_h \xrightarrow{\sim} \Gamma_{g,h} \circ U_{gh}\}$ is a set of natural isomorphisms of $\mathcal{C}$-module functors $\mathcal{M} \to (\mathcal{M}^g)^h$, i.e. the following commutes for $g, h \in G$, $X \in \mathcal{C}$, $M \in \mathcal{M}$:

$$\begin{tikzcd}
U_g^h(U_h(X \otimes^{\mathcal{M}} M)) \arrow[r, "U_g^h(\tau_h(X,M))"] \arrow[d, "\mu_{g,h}(X\otimes M)"] & U_g^h(X \otimes^h U_h(M)) \arrow[r, "\tau_g^h(X, U_h(M))"] & X \otimes^{h^g} U_g^h(U_h(M)) \arrow[d, "\mathrm{id}\otimes \mu_{g,h}(M)"] \\
\Gamma_{g,h}(U_{gh}(X \otimes^{\mathcal{M}} M)) \arrow[r, "\Gamma_{g,h}(\tau_{gh}(X,M))"'] & \Gamma_{g,h}(X \otimes^{gh} U_{gh}(M)) \arrow[r, "\Gamma_{g,h}(X, U_{gh}(M))"'] & X \otimes^{h^g} \Gamma_{g,h}(U_{gh}(M))
\end{tikzcd}$$

such that these satisfy a "compatibility condition" explained as follows: Notice that if we twist $\Gamma_{h,k} : \mathcal{M}^{hk} \xrightarrow{\sim} (\mathcal{M}^h)^k$ by g, we get a map $\Gamma_{h,k}^g : (\mathcal{M}^{hk})^g \xrightarrow{\sim} ((\mathcal{M}^h)^k)^g$. If we wanted to apply $\Gamma_{h,k}$ to the "outside", to get a $\mathcal{C}$-module equivalence $(\mathcal{M}^g)^{hk} \xrightarrow{\sim} ((\mathcal{M}^g)^h)^k$, we could do so by ${}^g\Gamma_{h,k} = (\mathrm{id}_{\mathcal{M}}, {}^g\Gamma_{h,k})$ where we define ${}^g\Gamma_{h,k}(X, M)$ as

$$X \otimes^{g^{(hk)}} M = T_g(T_{hk}(X)) \otimes^{\mathcal{M}} M \xrightarrow{T_g(\gamma_{h,k}(X)^{-1})\otimes \mathrm{id}} T_g(T_h(T_k(X))) \otimes^{\mathcal{M}} M = X \otimes^{g^{h^k}} M$$

Then we have that the following commutes via naturality of τ_g:

$$\begin{tikzcd}
U_g^{hk}(T_{hk}(X) \otimes^{\mathcal{M}} M) = U_g^{h^k}(\Gamma_{h,k}(X \otimes^{hk} M)) = U_g^{h^k}(T_{hk}(X) \otimes^{\mathcal{M}} M) \arrow[r, "U_g^{h^k}(\gamma_{h,k}(X)^{-1}\otimes \mathrm{id})"] \arrow[d, "\tau_g(T_{hk}(X), M)"] & U_g^{h^k}(T_h(T_k(X)) \otimes^{\mathcal{M}} M) \arrow[d, "\tau_g(T_h(T_k(X)), M)"] \\
T_{hk}(X) \otimes^g U_g^{hk}(M) = T_g(T_{hk}(X)) \otimes^{\mathcal{M}} U_g^{hk}(M) \arrow[r, "T_g(\gamma_{h,k}(X)^{-1}\otimes \mathrm{id})"] & T_g(T_h(T_k(X))) \otimes^{\mathcal{M}} U_g^{hk}(M) = T_h(T_k) \otimes^g U_g^{h^k}(M)
\end{tikzcd}$$

This diagram exactly expresses that $U_g^{h^k} \circ \Gamma_{h,k} = {}^g\Gamma_{h,k} \circ U_g^{hk}$ as $\mathcal{C}$-module functors. The compatibility condition is that the following holds:

$$\Gamma_{g,h}^k(\mu_{gh,k}) \circ \mu_{g,h}^k(U_k) = {}^g\Gamma_{h,k}(\mu_{g,hk}) \circ U_g^{h^k}(\mu_{h,k})$$

In terms of components, that means that for $M \in \mathcal{M}$ the following commutes:

$$\begin{tikzcd}
U_g(U_h(U_k(M))) = U_g^{h^k}(U_h^k(U_k(M))) \arrow[r, "\mu^k_{g,h}(U_k(M))"] \arrow[d, "U_g^{h^k}(\mu_{h,k}(M))"] & \Gamma^k_{g,h}(U^k_{gh}(U_k(M))) \arrow[d, "\Gamma^k_{g,h}(\mu_{gh,k})"] \\
U_g^{h^k}(\Gamma_{h,k}(U_{hk}(M))) = {}^g\Gamma_{h,k}(U_g^{hk}(U_{hk}(M))) \arrow[r, "{}^g\Gamma_{h,k}(\mu_{g,hk})"] & {}^g\Gamma_{h,k}(\Gamma_{g,hk}(U_{ghk}(M))) = U_{ghk}(M) = \Gamma^k_{g,h}(\Gamma_{gh,k}(U_{ghk}))
\end{tikzcd}$$

Example 2.8. Let us again consider the case when $\mathcal{C} = \mathrm{Vec}$. Recall that for a finite group G, the only action (T, γ) we have is trivial. Thus since $T_g = \mathrm{Id}_{\mathcal{C}}$, we have that $\mathcal{M}^g = \mathcal{M}$ as Vec-module categories for any Vec-module category $\mathcal{M}$. Since $\gamma_{g,h} = \mathrm{Id}_{\mathrm{Id}_{\mathrm{Vec}}}$ we have that $\Gamma_{g,h} = \mathrm{Id}_{\mathcal{M}}$ as a Vec-module functor. Let $\mathcal{M} = \mathrm{Vec}$, with action via the tensor product, and let's consider what the data of an G-equivariant Vec-module would be.

For each $g \in G$, we have $U_g : \mathrm{Vec} \to \mathrm{Vec}$, a Vec-module equivalence. In particular, it is then an equivalence of abelian categories, and hence trivial. So $U_g = \mathrm{Id}_{\mathrm{Vec}}$ as a functor, but not necessarily as a Vec-module functor. By the triangle axiom for module functors and strictness we get that $\tau_g(\Bbbk, V) = \mathrm{id}_V$. Then by strictness and the pentagon axiom for module functors, given $U, V \in \mathrm{Vec}$ we have the following:

$$\begin{aligned}
\tau_g(U,V) =& (\mathrm{id}_{U\otimes V}) \circ \tau_g(U,V) \\
=& (\mathrm{id}_U \otimes \mathrm{id}_V) \circ \tau_g(U,V) \\
=& (\mathrm{id}_U \otimes \tau_g(V,\Bbbk)) \circ \tau_g(U, V \otimes \Bbbk) \\
=& \tau_g(U \otimes V, \Bbbk) \\
=& \,\mathrm{id}_{U\otimes V}
\end{aligned}$$

Thus in fact we have that τ_g is trivial, and hence $U_g = \mathrm{Id}_{\mathrm{Vec}}$ as Vec-module functors.

We now have that $\mu_{g,h}$ is a natural isomorphism $\mathrm{Id}_{\mathrm{Vec}} \cong \mathrm{Id}_{\mathrm{Vec}}$ of Vec-module functors. Naturality gives us that $\mu_{g,h}(U \otimes V) = \mathrm{id}_U \otimes \mu_{g,h}(V) : U \otimes V \to U \otimes V$. Thus we have:

$$\mu_{g,h}(V) = \mu_{g,h}(V \otimes \Bbbk) = \mathrm{id}_V \otimes \mu_{g,h}(\Bbbk)$$

So $\mu_{g,h}$ is determined by $\mu_{g,h} : \Bbbk \xrightarrow{\sim} \Bbbk$. Then using the compatibility condition we get:

$$\mu_{gh,k}(\Bbbk) \circ \mu_{g,h}(\Bbbk) = \mu_{g,hk}(\Bbbk) \circ \mu_{h,k}(\Bbbk)$$

Recall that $\mathrm{Hom}_{\mathrm{Vec}}(\Bbbk, \Bbbk) \cong \Bbbk$, and $\mu_{g,l}(\Bbbk)$ is invertible, so we can regard it as an element of $\Bbbk^\times$. Thus μ gives us a function $\mu : G \times G \to \Bbbk^\times$ by $\mu(g,h) = \mu_{g,h}$. The above equation then expresses exactly that $\mu \in Z^2(G, \Bbbk^\times)$, i.e. that it's a 2-cocycle.

Definition 2.9. A *G-equivariant object* in a G-equivariant module category $(\mathcal{M}, U, \mu)$ is a pair (M, v), where $M \in \mathcal{M}$ and $v = \{u_g : U_g(M) \xrightarrow{\sim} M\}$ is a set of isomorphisms in $\mathcal{M}$ such that the following commutes for all $g, h \in G$:

$$\begin{tikzcd} U_g^h(U_h(M)) = U_g(U_h(M)) \arrow[r, "U_g(v_h)"] \arrow[d, "\mu_{g,h}(M)"] & U_g(M) \arrow[d, "v_g"] \\ \Gamma_{g,h}(U_{gh}(M)) = U_{gh}(M) \arrow[r, "v_{gh}"] & M \end{tikzcd}$$

Let $H \leq G$ be a subgroup, then H acts on $\mathcal{C}$ by simply restricting the action from G. So then we can talk about H-equivariant objects in an H-equivariant $\mathcal{C}$-module category $(\mathcal{M}, U, \mu)$. We define $\mathcal{M}^H$, the *equivariantization of* $\mathcal{M}$, to be the category of H-equivariant objects in $\mathcal{M}$, with a morphism $f : (M, v) \to (N, w)$, called an *H-equivariant map*, being a morphism $f : M \to N$ in $\mathcal{M}$ such that for all $h \in H$, the following commutes:

$$\begin{tikzcd} U_h(M) \arrow[r, "U_h(f)"] \arrow[d, "v_h"] & U_h(N) \arrow[d, "w_h"] \\ M \arrow[r, "f"] & N \end{tikzcd}$$

Example 2.10. Let us continue with the last example. So let $\mathcal{C} = \mathrm{Vec}$ with trivial action from G, $\mathcal{M} = \mathrm{Vec}$ with the tensor product action, and 2-cocycle μ. For $H \leq G$, an object $(W, \lambda) \in \mathcal{M}^H$ is a vector space along with isomorphisms $\lambda_h : W \xrightarrow{\sim} W$ such that for all $h, l \in H$ we have:

$$\lambda_h \circ \lambda_l = \lambda_{hl} \circ \mu_{g,h}(W) = (\lambda_{hl} \otimes \mathrm{id}_\Bbbk) \circ (\mathrm{id}_W \otimes \mu_{h,k}(\Bbbk)) = \mu(h,l)\lambda_{hl}$$

Thus an element of $\mathcal{M}^H$ is just a projective representation of H with Schur multiplier μ. A morphism $f : (W, \lambda) \to (L, \pi)$ in $\mathcal{M}^H$ is then just a linear map $f : W \to L$ such that for all $h \in H$

$$f \circ \lambda_h = \pi_h \circ f$$

In other words, it is an intertwining operator, or a homomorphism of representations. Thus $\mathcal{M}^H = \mathrm{Rep}_\mu(H)$, the category of H-representations with Schur multiplier μ.

Remark 2.11. A projective representation of H with Schur multiplier μ is the same as a representation of the twisted group algebra $\Bbbk^{\mu}[H]$.

Let $\mathcal{C}$ be a fusion category, (T, γ) be an action of a finite group G on $\mathcal{C}$, $H \leq G$, and $\mathcal{M}$ a H-equivariant $\mathcal{C}$-module category.

Theorem 2.12. *We have that $\mathcal{M}^H$ is a $\mathcal{C}^G$-module category with the following action:*

$$(X, u) \otimes^H (M, v) := (X \otimes^{\mathcal{M}} M, u \otimes v);$$

$$[u \otimes v]_h := U_h(X \otimes^{\mathcal{M}} M) \xrightarrow{\tau_h(X,M)} T_h(X) \otimes^{\mathcal{M}} U_h(M) \xrightarrow{v_h \otimes u_h} X \otimes^{\mathcal{M}} M$$

Proof. To see that $u \otimes v$ is an equivariant structure on $X \otimes^{\mathcal{M}} M$, refer to the below diagram:

$$\begin{tikzcd}
U_h^k(U_k(X \otimes^{\mathcal{M}} M)) = U_h(U_k(X \otimes^{\mathcal{M}} M)) \arrow[r, "U_h(\tau_{h,k}(X,M))"] \arrow[dd, "\mu_{h,k}(X,M)"] & U_h(X \otimes^k U_k(M)) = U_h(T_k(X) \otimes^{\mathcal{M}} U_k(M)) \arrow[r, "U_h(u_k \otimes v_k)"] \arrow[d, "\tau_h(X, U_k(M))"] & U_h(X \otimes^{\mathcal{M}} M) \arrow[d, "\tau_h(X,M)"] \\
 & T_h(T_k(X)) \otimes^{\mathcal{M}} U_h(U_k(M)) \arrow[r, "T_h(u_k) \otimes U_h(v_k)"] \arrow[d, "\gamma_{h,k}(X) \otimes \mu_{h,k}(M)"] & T_h(X) \otimes U_h(M) \arrow[d, "u_k \otimes v_k"] \\
\Gamma_{h,k}(U_{hk}(X \otimes^{\mathcal{M}} M)) = U_{hk}(X \otimes^{\mathcal{M}} M) \arrow[r, "\tau_{hk}(X,M)"] & X \otimes^{hk} U_{hk}(M) = T_{hk}(X) \otimes^{\mathcal{M}} U_{hk}(M) \arrow[r, "u_{hk} \otimes v_{hk}"] & X \otimes^{\mathcal{M}} M
\end{tikzcd}$$

Notice that the perimeter of this diagram is exactly the diagram for equivariance, so we show that it commutes. The lower right square commutes by equivariance of X and M tensored together. The upper right square commutes via naturality of τ. Finally, the left pentagon commutes due to naturality of μ since the lower right arrow can be broken into $(\gamma_{h,k}(X) \otimes \mathrm{id}) \circ (\mathrm{id} \otimes \mu_{h,k})$, and then one can invert $\gamma_{h,k}(X) \otimes \mathrm{id}$ to recover the diagram for naturality of μ in Definition 2.7.

Note that the unit object in $\mathcal{C}^G$ is $(1, \mathrm{id})$, where $\mathrm{id}_g = \mathrm{id}_{T_g(1)} = \mathrm{id}_1$. We will define associativity and unit constraints as follows:

$$m^H((X, u), (Y, w), (M, v)) := m(X, Y, M)$$

$$l^H((M, v)) := l(M)$$

First, we will show that these are both well defined, i.e. morphisms in $\mathcal{M}^H$. For l^H, refer to the below diagram, whose perimeter is the diagram for l^H to be a morphism in $\mathcal{M}^H$:

$$\begin{tikzcd}
U_h(1 \otimes^{\mathcal{M}} M) \arrow[r, "\tau_h(1,M)"] \arrow[d, "U_h(l(M))"] & 1 \otimes^h U_h(M) = 1 \otimes^{\mathcal{M}} U_h(M) \arrow[r, "\mathrm{id} \otimes v_h"] \arrow[dl, "l^H(U_h(M)) = l(U_h(M))"] & 1 \otimes^{\mathcal{M}} M \arrow[d, "l(M)"] \\
U_h(M) \arrow[rr, "v_h"'] & & M
\end{tikzcd}$$

The left triangle commutes by the fact that U_h is a $\mathcal{C}$-module functor (second diagram in Definition 1.37), and the right quadrilateral commutes via naturality of l. Thus l^H is a morphism in $\mathcal{M}^H$. For m^H, refer to the below diagram, whose perimeter expresses that m^H is in $\mathcal{M}^H$:

$$\begin{tikzcd}
U_h((X \otimes Y) \otimes^{\mathcal{M}} M) \arrow[r, "\tau_h(X \otimes Y, M)"] \arrow[d, "U_h(m(X,Y,M))"] & T_h(X \otimes Y) \otimes^{\mathcal{M}} U_h(M) \arrow[r, "t_h^{-1}(X,Y) \otimes \mathrm{id}"] \arrow[dr, "m^h(X,Y,U_h(M))"'] & (T_h(X) \otimes T_h(Y)) \otimes^{\mathcal{M}} U_h(M) \arrow[r, "(u_h \otimes w_h) \otimes v_h"] \arrow[d, "m(T_h(Y),T_h(X),U_h(M))"] & (X \otimes Y) \otimes^{\mathcal{M}} M \arrow[d, "m(X,Y,M)"] \\
U_h(X \otimes^{\mathcal{M}} (Y \otimes^{\mathcal{M}} M)) \arrow[r, "\tau_h(X, Y \otimes^{\mathcal{M}} M)"] & T_h(X) \otimes^{\mathcal{M}} U_h(Y \otimes^{\mathcal{M}} M) \arrow[r, "\mathrm{id} \otimes \tau_h(Y,M)"] & T_h(X) \otimes^{\mathcal{M}} (T_h(Y) \otimes^{\mathcal{M}} M) \arrow[r, "u_h \otimes (w_h \otimes v_h)"] & X \otimes^{\mathcal{M}} (Y \otimes^{\mathcal{M}} M)
\end{tikzcd}$$

The left pentagon commutes since (U_h, τ_h) is a $\mathcal{C}$-module functor, the middle triangle commutes by definition of m^h, and the right square commutes via naturality of m. Thus m^H is a morphism in $\mathcal{M}^H$. That m^H and l^H satisfy the diagrams in Definition 1.35 follows immediately from the fact that m and l do since $\mathcal{M}$ is a $\mathcal{C}$-module category. $\square$

2.3 Classification of Module Categories over the Equivariantization of a Fusion Category

For the purpose of this section we assume that $\mathcal{C}$ is a fusion category acted on by a finite group G with action T.

Definition 2.13. A $\mathcal{C}$-module category $\mathcal{M}$ is called *indecomposable* if it cannot be written as a direct sum of nontrivial $\mathcal{C}$-module categories.

Theorem 2.14. (Etingof et al., 2011, Proposition 5.4) *Every indecomposable $\mathcal{C}^G$-module category is equivalent to some $\mathcal{M}^H$ where $H \leq G$ and $\mathcal{M}$ is an indecomposable H-equivariant $\mathcal{C}$-module category.*

We will give a brief proof sketch of the following classification theorem since the relevant definitions have been changed. However, before this, we will record some relevant definitions and results that are required.

Definition 2.15. Let Vec_G be the category of G-graded finite dimensional vector spaces. This will be a fusion category, and by abuse of notation we can denote its simple objects by $g \in G$ where $g \in \mathrm{Vec}_G$ is the vector space with $\Bbbk$ in the g-component and nothing else.

Definition 2.16. We define the *crossed product category* $\mathcal{C} \rtimes G$ as follows. It is a semisimple $\Bbbk$-linear abelian category with simple objects written as $X \boxtimes g$ where $X \in \mathcal{C}$ is simple, and $g \in G$. We define the hom-spaces by $\mathrm{Hom}_{\mathcal{C} \rtimes G}(X \boxtimes g, Y \boxtimes h) := \mathrm{Hom}_{\mathcal{C}}(X, Y) \otimes \mathrm{Hom}_{\mathrm{Vec}_G}(g, h)$. We give this a tensor structure as follows:

$$(X \boxtimes g) \otimes (Y \boxtimes h) := (X \otimes T_g(Y)) \boxtimes gh$$

This will have unit object $1 \boxtimes e$ (where $e \in G$ is the unit), and one can write out the appropriate constraints. This will be a fusion category since $\mathcal{C}$ is.

Remark 2.17. $\mathcal{C}$ can be found as a tensor subcategory of $\mathcal{C} \rtimes G$ by taking $\mathcal{C} \boxtimes e$.

$\mathcal{C}$ can be demonstrated as an exact left $\mathcal{C} \rtimes G$ module category with the following action:

$$(C \rtimes G) \times \mathcal{C} \to \mathcal{C}; (Y \boxtimes g) \otimes X := T_g(X) \otimes Y^*$$

Remark 2.18. This left action can be seen as coming from the more obvious right action $X \otimes (Y \boxtimes g) := T_{g^{-1}}(X \boxtimes Y)$ in section 3.1 of Nikshych (2008) by using the dual.

Definition 2.19. Let $\mathcal{M}, \mathcal{N}$ be $\mathcal{C}$-module categories.

1. Let $\mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{N})$ denote the category of right exact $\mathcal{C}$-module functors and $\mathcal{C}$-module natural transformations between them. This will be a finite abelian category.

2. Denote $\mathrm{Fun}_{\mathcal{C}}(\mathcal{M}, \mathcal{M})$ by $\mathcal{C}^*_{\mathcal{M}}$ and call it the *dual category of* $\mathcal{C}$ *with respect to* $\mathcal{M}$. This will be fusion if $\mathcal{M}$ is indecomposable. $\mathcal{M}$ will be a left $\mathcal{C}^*_{\mathcal{M}}$ module.

3. If $\mathcal{D}$ is another fusion category, we say that $\mathcal{C}$ and $\mathcal{D}$ are *categorically Morita equivalent* if there is an exact $\mathcal{C}$ module category $\mathcal{M}$ such that $\mathcal{D}^{\mathrm{op}} \cong \mathcal{C}^*_{\mathcal{M}}$ (note this is the monoidal opposite, see Remark 1.13).

Theorem 2.20. *$\mathcal{C}^G$ and $\mathcal{C} \rtimes G$ are categorically Morita equivalent via $\mathcal{C}$ as a $\mathcal{C} \rtimes G$-module category.*

Remark 2.21. The above is constructed by sending (X, u) to $X^* \otimes -$. The $\mathcal{C} \rtimes G$-module functor structure is given using u as follows:

$$\begin{aligned} J(Y,Z) : F_X((Y \boxtimes g) \otimes Z) =& X^* \otimes (T_g(Z) \otimes Y^*) \\ \xrightarrow{(u_g^{-1})^* \otimes \mathrm{id} \otimes \mathrm{id}}& T_g(X^*) \otimes T_g(Z) \otimes Y \\ \xrightarrow{t_g(X^*, Z) \otimes \mathrm{id}}& T_g(X^* \otimes Z) \otimes Y \\ =& (Y \boxtimes g) \otimes (X^* \otimes Z) \\ =& (Y \boxtimes g) \otimes F_X(Z) \end{aligned}$$

Corollary 2.22. Due to Theorem 7.12.16 in Etingof et al. (2015), this allows us to conclude that there is an equivalence of 2-categories as below:

$$\mathrm{Mod}(\mathcal{C} \rtimes G) \to \mathrm{Mod}(\mathcal{C}^G); \mathcal{N} \mapsto \mathrm{Fun}_{\mathcal{C} \rtimes \mathcal{G}}(\mathcal{C}, \mathcal{N})$$

Proof Sketch of Theorem 2.14. Let $\mathcal{N}$ be an indecomposable $\mathcal{C} \rtimes G$ module category. Then since $\mathcal{C}$ can be regarded as a tensor subcategory of $\mathcal{C} \rtimes G$, $\mathcal{N}$ is naturally also a $\mathcal{C}$-module category. However, $\mathcal{N}$ may not be indecomposable over $\mathcal{C}$, so write its decomposition:

$$\mathcal{N} = \bigoplus_{s \in S} \mathcal{N}_s$$

For $g \in G$, define $\mathcal{N}_{gs} = (1 \boxtimes g^{-1}) \otimes \mathcal{N}_s$. This will be an indecomposable $\mathcal{C}$-module subcategory of $\mathcal{N}$. This yields a (right) action of G on S which is transitive since $\mathcal{N}$ is indecomposable over $\mathcal{C} \rtimes G$. Then we can define a $\mathcal{C}$-module equivalence $(U_g, \tau_g) : \mathcal{N}_{gs} \to (\mathcal{N}_s)^g$ where

$U_g(N) = (1 \boxtimes g) \otimes N$, and τ_g is defined as below with n being the module structure of $\mathcal{N}$:

$$\begin{aligned}
\tau_g(X,N) : U_g(X \otimes N) =& (1 \boxtimes g) \otimes [(X \boxtimes e) \otimes N] \\
\xrightarrow{n^{-1}((1\boxtimes g),(X\boxtimes e),N)} & [(1 \boxtimes g) \otimes (X \boxtimes e)] \otimes M \\
=& [T_g(X) \boxtimes g] \otimes N \\
=& [(T_g(X) \boxtimes e) \otimes (1 \boxtimes g)] \otimes N \\
\xrightarrow{n((T_g(X)\boxtimes e),(1\boxtimes g),N)} & (T_g(X) \boxtimes e) \otimes [(1 \boxtimes g) \otimes N] \\
=& X \otimes^g U_g(X)
\end{aligned}$$

Now, let $s \in S$ and let $H := \mathrm{Stab}(s)$. Then $S \cong G/H$. Set $\mathcal{M} = \mathcal{N}_s$ so that for any $h \in H$, we have that $\mathcal{M} = \mathcal{N}_s = \mathcal{N}_{hs} \cong (\mathcal{N}_s)^h = \mathcal{M}^h$ by (U_h, τ_h). We can then define a natural transformation of $\mathcal{C}$-module functors, $\mu_{h,k} : (U_h)^k \circ U_k \xrightarrow{\sim} \Gamma_{h,k} \circ U_{hk}$, using the module structure of $\mathcal{M}$ as defined below:

$$\begin{aligned}
[(U_h)^k \circ U_k](M) =& U_h^k(U_k(M)) \\
=& (1 \boxtimes h) \otimes [(1 \boxtimes k) \otimes M] \\
\xrightarrow{n^{-1}((1\boxtimes h),(1\boxtimes k),M)} & [(1 \boxtimes h) \otimes (1 \boxtimes k)] \otimes M \\
=& (1 \boxtimes hk) \otimes M \\
=& U_{hk}(M) \\
=& [\Gamma_{h,k} \circ U_{hk}](M)
\end{aligned}$$

One can check that these satisfy the compatibility conditions in Definition 2.7. Thus $(\mathcal{M}, U, \mu)$ defines an H-equivariant $\mathcal{C}$-module category.

Conversely, suppose we are given some $H \leq G$ and H-equivariant $\mathcal{C}$-module category $\mathcal{M}$. We can then define $S := G/H$ fixing some $s \in S$ and regarding H as $\mathrm{Stab}(s)$ with respect to the obvious transitive action of G on S. For $t \in S$, let $\mathcal{N}_t$ formally have elements $(1 \boxtimes g^{-1}) \otimes M$ where $M \in \mathcal{M}$ and $g \in G$ is such that $t = gs$. We then define a $\mathcal{C} \rtimes G$ module action on $\mathcal{N} := \bigoplus_{t \in S} \mathcal{N}_t$:

$$(X \boxtimes h) \otimes [(1 \boxtimes g^{-1}) \otimes M] = (1 \boxtimes hg^{-1}) \otimes [T_{gh^{-1}}(X) \otimes M]$$

If we were given $\mathcal{N}$, this will return something equivalent, similarly using this $\mathcal{N}$ to form H and $\mathcal{M}$ will return something equivalent. So H and $\mathcal{M}$ determine $\mathcal{N}$ and vice versa (up to equivalence). Recall from Corollary 2.22 that we have an equivalence of the categories of module categories between $\mathcal{C} \rtimes G$ and $\mathcal{C}^G$. One can show that $\mathcal{M}^H$ is equivalent as a $\mathcal{C}^G$ module category to $\mathrm{Fun}_{\mathcal{C}\rtimes G}(\mathcal{C}, \mathcal{N})$. The $\mathcal{M}^H \to \mathrm{Fun}_{\mathcal{C}\rtimes G}(\mathcal{C}, \mathcal{N})$ half of this equivalence is by sending $(M, v) \in \mathcal{M}^H$ to $(F_{M,v}, s_{M,v}) \in \mathrm{Fun}_{\mathcal{C}\rtimes G}(\mathcal{C}, \mathcal{N})$ where $F_{M,v}(X) = (X^* \boxtimes e) \otimes M$ and $s_{M,v}$ is given by

$$\begin{aligned}
s_{M,v}((Y \boxtimes g), Y) : F_{M,v}((Y \boxtimes g) \otimes X) =& F_{M,v}(T_g(X) \otimes Y^*) \\
=& [(T_g(X) \otimes Y^*)^* \boxtimes e] \otimes M \\
=& [(Y \otimes T_g(X^*)) \boxtimes e] \otimes M \\
=& [(Y \boxtimes g) \otimes (X^* \boxtimes g^{-1})] \otimes M \\
\xrightarrow{n((Y\boxtimes g),(X^*\boxtimes g^{-1}),M)}& (Y \boxtimes g) \otimes [(X^* \boxtimes g^{-1}) \otimes M] \\
\xrightarrow{\mathrm{id} \otimes \mathrm{id} \otimes v_g^{-1}}& (Y \boxtimes g) \otimes [(X^* \boxtimes g^{-1}) \otimes ((1 \boxtimes g) \otimes M)] \\
\xrightarrow{\mathrm{id} \otimes n^{-1}((X^*\boxtimes g^{-1}),(1\boxtimes g),M)}& (Y \boxtimes g) \otimes ([(X^* \boxtimes g^{-1}) \otimes (1 \boxtimes g)] \otimes M) \\
=& (Y \boxtimes g) \otimes [(X^* \boxtimes e) \otimes M] \\
=& (Y \boxtimes g) \otimes F_{M,v}(X)
\end{aligned}$$

The other half is given by sending $(F, l) \in \mathrm{Fun}_{\mathcal{C}\rtimes G}(\mathcal{C}, \mathcal{N})$ to $(M_{F,l}, v_{F,l}) \in \mathcal{M}^H$ as follows: Notice that $F(1)$ will be simple in $\mathcal{N}$, so $F(1) \in \mathcal{N}_t$ for some $t \in S$. Let ℓ be such that $\ell t = s$. Then define $M_{F,l} := (1 \boxtimes \ell) \otimes F(1) \in \mathcal{N}_{\ell t} = \mathcal{N}_s = \mathcal{M}$, and $v_{F,l}$ as below:

$$\begin{aligned}
(v_{F,l})_h : (1 \boxtimes h) \otimes M =& (1 \boxtimes h) \otimes ((1 \boxtimes \ell) \otimes F(1)) \\
\xrightarrow{n^{-1}((1\otimes h),(1\otimes \ell),F(1))}& (1 \boxtimes h\ell) \otimes F(1) \\
\xrightarrow{l^{-1}((1\boxtimes h\ell),1)}& F((1 \boxtimes h\ell) \otimes 1) \\
=& F(1) \\
=& F((1 \boxtimes \ell) \otimes 1) \\
\xrightarrow{l((1\boxtimes \ell),1)}& (1 \boxtimes \ell) \otimes F(1) = M
\end{aligned}$$

With this equivalence in hand, the proof is completed since by the 2-equivalence in Corollary 2.22, any indecomposable $\mathcal{C}^G$ module category $\mathcal{L}$ is equivalent to one of the form $\mathrm{Fun}_{\mathcal{C} \rtimes G}$ for an indecomposable $\mathcal{C} \rtimes G$ module category $\mathcal{N}$ which determines $H \leq G$ and an H-equivariant $\mathcal{C}$-module category $\mathcal{M}$ such that $\mathrm{Fun}_{\mathcal{C} \rtimes G}(\mathcal{C}, \mathcal{N})$ is equivalent to $\mathcal{M}^H$. $\square$

CHAPTER 3. INTERNAL HOMS OF A MODULE CATEGORY ACTION

In category theory, there is sometimes a dichotomy between "internal" and "external" structure. This refers to information which is available to the category versus information that is merely part of its structure. Take, for example, the collections $\mathrm{Hom}_{\mathcal{C}}(X, Y)$ for $X, Y \in \mathcal{C}$. In most cases, one cannot reason about these collections and remain inside the category we are studying since these are no longer objects of our category. For example, in the free arrow category $\cdot \to \cdot$, the hom sets are clearly not objects of this category. Contrast this to Vec. The set of linear maps between two vector spaces is itself a vector space. It's not just that we have given the set a vector space structure, but the vector space structure is actually meaningful in some way. We can see this via the "tensor-hom" adjunction for vector spaces:

$$\mathrm{Hom}_{\mathrm{Vec}}(U \otimes V, W) \cong \mathrm{Hom}_{\mathrm{Vec}}(U, \mathrm{Hom}_{\mathrm{Vec}}(V, W))$$

In general for a monoidal category $\mathcal{C}$, we say that it admits internal homs if for each $Y \in \mathcal{C}$ the functor $- \otimes Y$ has a right adjoint $[Y, -]$. In other words, we have

$$\mathrm{Hom}_{\mathcal{C}}(X \otimes Y, Z) \cong \mathrm{Hom}_{\mathcal{C}}(X, [Y, Z])$$

For further exposition of these, see Eilenberg and Kelly (1966). Now, revisiting Remark 1.18, we clearly see that any rigid monoidal category has internal homs with $[Y, -] = - \otimes Y^*$.

Now, we generalize this notion to module categories over a fusion category. In the case where we consider $\mathcal{C}$ as a $\mathcal{C}$-module category by the tensor product, this will return the more familiar notion above. Let $\mathcal{M}$ be a $\mathcal{C}$-module category. If $M, N \in \mathcal{M}$, then we get a functor:

$$\mathrm{Hom}_{\mathcal{M}}(- \otimes^{\mathcal{M}} M, N) : \mathcal{C} \to \mathrm{Vec}$$

Since the module product bifunctor is biexact, and hom functors are left exact, the above functor is left exact. Thus it is representable (see Etingof et al. (2015) Corollary 1.8.11).

Definition 3.1. The representing object for the above functor is called an *internal hom*, and is denoted $\underline{\mathrm{Hom}}(M, N)$. In other words, we have a natural isomorphism:

$$\mathrm{Hom}_{\mathcal{M}}(X \otimes^{\mathcal{M}} M, N) \cong \mathrm{Hom}_{\mathcal{C}}(X, \underline{\mathrm{Hom}}(M, N))$$

3.1 Modules over an Algebra in a Fusion Category

We will define the notion of an algebra, and modules over that algebra, internal to a fusion category $\mathcal{C}$. The categories of internal right modules of an internal algebra are important to studying the module categories of $\mathcal{C}$ in the following way:

Theorem 3.2. (Etingof et al., 2015, Theorem 7.10.1) *Let $\mathcal{M}$ be an indecomposable $\mathcal{C}$-module category. Then there is an algebra A internal to $\mathcal{C}$ such that $\mathcal{M}$ is equivalent, as a $\mathcal{C}$-module category, to the category of internal right A- modules.*

Remark 3.3. One of the uses of internal homs with respect to module category actions is that the above A is of the form $\underline{\mathrm{Hom}}(M, M)$ for $M \in \mathcal{M}$.

The definitions of algebras, algebra homomorphisms, modules, and module homomorphisms internal to $\mathcal{C}$ are what one would expect. We simply write down the relevant structure maps and the commutative diagrams one expects them to satisfy.

Definition 3.4. Let $\mathcal{C}$ be a fusion category. Then an *algebra* in $\mathcal{C}$ is a triple (A, m, u) where $A \in \mathcal{C}$, and we have a multiplication morphism $m : A \otimes A \to A$, and a unit morphism $u : 1 \to A$ such that the following commute, expressing the associativity and unitality of A:

$$\begin{tikzcd} A \otimes A \otimes A \arrow[r, "\mathrm{id} \otimes m"] \arrow[d, "m \otimes \mathrm{id}"] & A \otimes A \arrow[d, "m"] \\ A \otimes A \arrow[r, "m"] & A \end{tikzcd} \qquad \begin{tikzcd} 1 \otimes A \arrow[r, "u \otimes \mathrm{id}"] \arrow[dr, "l_A"'] & A \otimes A \arrow[d, "m"] & A \otimes 1 \arrow[l, "\mathrm{id} \otimes u"'] \arrow[dl, "r_A"] \\ & A & \end{tikzcd}$$

Definition 3.5. Let $\mathcal{C}$ be a fusion category, and (A, m, u) an algebra in $\mathcal{C}$. A *left A-module* in $\mathcal{C}$ is a pair (M, p) where $M \in \mathcal{C}$ and $p : A \otimes M \to M$ is a morhpism in $\mathcal{C}$ (called the action of A on M) such that the following diagrams commute:

$$\begin{tikzcd} A \otimes A \otimes M \arrow[r, "\mathrm{id} \otimes p"] \arrow[d, "m\otimes \mathrm{id}"] & A \otimes M \arrow[d, "p"] \\ A \otimes M \arrow[r, "p"] & M \end{tikzcd} \qquad \begin{tikzcd} 1 \otimes M \arrow[r, "l_M"] \arrow[d, "\mathrm{id} \otimes u"] & M \\ A \otimes M \arrow[ur, "p"'] & \end{tikzcd}$$

A morphism $f : (M, p) \to (N, q)$ of left A-modules is a morphism $f : M \to N$ in $\mathcal{C}$ such that the following commutes:

$$\begin{tikzcd} A \otimes M \arrow[r, "\mathrm{id} \otimes f"] \arrow[d, "p"] & A \otimes N \arrow[d, "q"] \\ M \arrow[r, "f"] & N \end{tikzcd}$$

A right A-module (N, q) is defined similarly, with an action $q : A \otimes N \to N$.

Remark 3.6. We are omitting the associativity and unit constraints in the above definitions.

Definition 3.7. Given an algebra (A, m, u) in $\mathcal{C}$, a left module (M, p) and a right module (N, q), the *tensor product of* M *and* N *over* A, $M \otimes_A N$ is defined to be the coequalizer below:

$$M \otimes A \otimes N \underset{\mathrm{id} \otimes q}{\overset{p\otimes \mathrm{id}}{\rightrightarrows}} M \otimes N \xrightarrow{e} M \otimes_A N$$

Example 3.8. If $\mathcal{C} = \mathrm{Vec}$, algebras, modules, and tensor products are the usual ones.

Given any $X \in \mathcal{C}$, and a right A module (M, p) in $\mathcal{C}$, we can define an obvious module structure on $X \otimes M$ by $\mathrm{id} \otimes p$. Letting $\mathrm{Mod}_{\mathcal{C}}(A)$ be the category of right A-modules, we then get a functor:

$$\mathcal{C} \times \mathrm{Mod}_{\mathcal{C}}(A) \to \mathrm{Mod}_{\mathcal{C}}(A)$$

Naturality of the association constraint a and the left unit morphism l of $\mathcal{C}$ immediately give us that $a(X, Y, M)$ and $l(M)$ are maps of modules for $X, Y \in \mathcal{C}$, $M \in \mathrm{Mod}_{\mathcal{C}(A)}$. The pentagon and triangle axioms for $\mathcal{C}$ then give us that this functor turns $\mathrm{Mod}_{\mathcal{C}}(A)$ into a $\mathcal{C}$-module category.

We now will compute the internal homs with respect to this action. This was remarked in Ostrik (2003), but we record a proof of it here.

Lemma 3.9. Let A be an algebra in a fusion category $\mathcal{C}$ and (N, q) be a right A-module. Then $({}^*N, q')$ is a left A-module, with q' the image of q under

$$\mathrm{Hom}_{\mathcal{C}}(N \otimes A, N) \cong \mathrm{Hom}_{\mathcal{C}}(A, {}^*N \otimes N) \cong \mathrm{Hom}_{\mathcal{C}}(A \otimes {}^*N, {}^*N)$$

Proof. Following q through the first isomorphism we get

$$\tilde{q} : A \xrightarrow{\mathrm{coev}_N \otimes \mathrm{id}} {}^*N \otimes N \otimes A \xrightarrow{\mathrm{id} \otimes q} {}^*N \otimes N$$

Then sending $\tilde{q}$ through the second yields

$$q' : A \otimes {}^*N \xrightarrow{\tilde{q} \otimes \mathrm{id}} {}^*N \otimes N \otimes {}^*N \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_N} {}^*N$$

Putting these together we get the following:

$$q' : A \otimes {}^*N \xrightarrow{\mathrm{coev}_N \otimes \mathrm{id} \otimes \mathrm{id}} {}^*N \otimes N \otimes A \otimes {}^*N \xrightarrow{\mathrm{id} \otimes q \otimes \mathrm{id}} {}^*N \otimes N \otimes {}^*N \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_N} {}^*N$$

For the triangle in the definition of a left module, refer to the below diagram:

$$\begin{tikzcd}
1 \otimes {}^*N \arrow[r, "\mathrm{coev}_N \otimes \mathrm{id} \otimes \mathrm{id}"] \arrow[d, "u \otimes \mathrm{id}"] & {}^*N \otimes N \otimes 1 \otimes {}^*N \arrow[r, "\mathrm{id} \otimes r_N \otimes \mathrm{id}"] \arrow[d, "\mathrm{id} \otimes \mathrm{id} \otimes u \otimes \mathrm{id}"] & {}^*N \otimes N \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_N"] & {}^*N \\
A \otimes {}^*N \arrow[r, "\mathrm{coev}_N \otimes \mathrm{id} \otimes \mathrm{id}"] & {}^*N \otimes N \otimes A \otimes {}^*N \arrow[ur, "\mathrm{id} \otimes q \otimes \mathrm{id}"'] & &
\end{tikzcd}$$

The top path is $l_{^*N}$ since ${}^*(-)$ is a monoidal functor to the opposite category. The bottom path is $q' \circ (u \otimes \mathrm{id})$, so commutativity of this diagram gives the triangle in the definition of a left module commuting. The left square clearly commutes, and the triangle commutes since (N, q) is a right module. Now to see the square, refer to the diagram below, whose perimeter is the needed condition for associativity of q':

$$
\begin{tikzcd}
A \otimes A \otimes {}^*N \arrow[rr, "\mathrm{id} \otimes \mathrm{coev}_N \otimes \mathrm{id} \otimes \mathrm{id}"] \arrow[dddd, "m \otimes \mathrm{id}"] \arrow[dr, "\mathrm{coev}_N \otimes \mathrm{id}^{\otimes 3}"] & & A \otimes N^* \otimes N \otimes A \otimes N^* \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes q \otimes \mathrm{id}"] \arrow[d, "\mathrm{coev}_N \otimes \mathrm{id}^{\otimes 5}"] & A \otimes {}^*N \otimes N \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{ev}_N"] & A \otimes {}^*N \arrow[d, "\mathrm{coev}_N \otimes \mathrm{id} \otimes \mathrm{id}"] \\
 & {}^*N \otimes N \otimes A \otimes A \otimes {}^*N \arrow[r, "\mathrm{id}^{\otimes 3} \otimes \mathrm{coev}_N \otimes \mathrm{id}"] \arrow[d, "\mathrm{id} \otimes q \otimes \mathrm{id} \otimes \mathrm{id}"] \arrow[dddr, bend right, "\mathrm{id} \otimes \mathrm{id} \otimes m \otimes \mathrm{id}"] & {}^*N \otimes N \otimes A \otimes {}^*N \otimes N \otimes A \otimes {}^*N \arrow[d, "\mathrm{id} \otimes q \otimes \mathrm{id}^{\otimes 4}"] & & {}^*N \otimes N \otimes A \otimes {}^*N \arrow[d, "\mathrm{id} \otimes q \otimes \mathrm{id}"] \\
 & {}^*N \otimes N \otimes A \otimes {}^*N \arrow[r, "\mathrm{id}^{\otimes 2} \otimes \mathrm{coev}_N \otimes \mathrm{id}^{\otimes 2}"] \arrow[dr, equal] & {}^*N \otimes N \otimes {}^*N \otimes N \otimes A \otimes {}^*N \arrow[r, "\mathrm{id}^{\otimes 3} \otimes q \otimes \mathrm{id}"] \arrow[d, "\mathrm{id} \otimes \mathrm{ev}_N \otimes \mathrm{id}^{\otimes 3}"] & {}^*N \otimes N \otimes {}^*N \otimes N \otimes {}^*N \arrow[r, "\mathrm{id}^{\otimes 3} \otimes \mathrm{ev}_N"] \arrow[dd, "\mathrm{id} \otimes \mathrm{ev}_N \otimes \mathrm{id}^{\otimes 3}"] & {}^*N \otimes N \otimes {}^*N \arrow[dd, "\mathrm{id} \otimes \mathrm{ev}_N"] \\
 & & {}^*N \otimes N \otimes A \otimes {}^*N \arrow[dr, "\mathrm{id} \otimes q \otimes \mathrm{id}"] & & \\
A \otimes {}^*N \arrow[rr, "\mathrm{coev}_N \otimes \mathrm{id} \otimes \mathrm{id}"] & & {}^*N \otimes N \otimes A \otimes {}^*N \arrow[r, "\mathrm{id} \otimes q \otimes \mathrm{id}"] & {}^*N \otimes N \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_N"] & {}^*N
\end{tikzcd}
$$

The upper right rectangle, the two lower right quadrilaterals, the upper left quadrilateral, the central left rectangle, and the left most four sided shape all clearly commute. The central triangle commutes via the definition of duals. The lower central four sided shape commutes since (N, q) is a right A-module. Thus the above diagram commutes, and hence $({}^*N, q')$ is a left A-module, as desired. $\square$

Theorem 3.10. *Let A be an algebra in $\mathcal{C}$, a fusion category. Let (M, p) and (N, q) be right A-modules and $X \in \mathcal{C}$. Then there is an adjunction:*

$$\mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, X) \cong \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(A)}(M, X \otimes N)$$

Proof. Recall that $M \otimes_A {}^*N$ is defined as the coequalizer below:

$$\begin{tikzcd} M \otimes A \otimes N \arrow[r, shift left, "p\otimes\mathrm{id}"] \arrow[r, shift right, "\mathrm{id}\otimes q'"'] & M \otimes {}^*N \arrow[r, "e"] & M \otimes_A {}^*N \end{tikzcd}$$

Then we have the following composition:

$$\phi : \mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, X) \xrightarrow{\mathrm{Hom}(e,-)} \mathrm{Hom}_{\mathcal{C}}(M \otimes {}^*N, X) \cong \mathrm{Hom}_{\mathcal{C}}(M, X \otimes N)$$

We wish to show that the image of this map lies in $\mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(A)}(M, X \otimes N)$. To this end, let $f : M \otimes_A {}^*N \to X$. Then $\phi(f)$ is a map of modules iff the following commutes:

$$\begin{tikzcd} M \otimes A \arrow[r, "\phi(f)\otimes\mathrm{id}"] \arrow[d, "p"] & X \otimes N \otimes A \arrow[d, "\mathrm{id}\otimes q"] \\ M \arrow[r, "\phi(f)"] & X \otimes N \end{tikzcd}$$

To see that this commutes, refer to the following, whose perimeter is the same diagram as above:

$$\begin{tikzcd} M \otimes A \arrow[rr, "\mathrm{id}\otimes\mathrm{coev}_N\otimes\mathrm{id}"] \arrow[dr, "\mathrm{id}\otimes\mathrm{id}\otimes\mathrm{coev}_N"] \arrow[dd, "p"] & & M \otimes {}^*N \otimes N \otimes A \arrow[r, "e\otimes\mathrm{id}\otimes\mathrm{id}"] \arrow[dd, "\mathrm{id}\otimes\mathrm{id}\otimes q"] & (M \otimes_A {}^*N) \otimes N \otimes A \arrow[r, "f\otimes\mathrm{id}\otimes\mathrm{id}"] \arrow[dd, "\mathrm{id}\otimes\mathrm{id}\otimes q"] & X \otimes N \otimes A \arrow[dd, "\mathrm{id}\otimes q"] \\ & M \otimes A \otimes {}^*N \otimes N \arrow[dr, shift left, "\mathrm{id}\otimes q'\otimes\mathrm{id}"] \arrow[dr, shift right, "p\otimes\mathrm{id}\otimes\mathrm{id}"'] & & & \\ M \arrow[rr, "\mathrm{id}\otimes\mathrm{coev}_N"] & & M \otimes {}^*N \otimes N \arrow[r, "e\otimes\mathrm{id}"] & (M \otimes_A {}^*N) \otimes N \arrow[r, "f\otimes\mathrm{id}"] & X \otimes N \end{tikzcd}$$

Suppose each quadrilateral commutes individually, since the double arrows followed by $e \otimes \mathrm{id}$ are a coequalizer diagram tensored with identity, the perimeter commutes. Thus we would have that ϕ is a map $\mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, X) \to \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(\mathcal{A})}(M, X \otimes N)$.

It is clear that the right two squares and the bottom left quadrilateral commute, so we will show that the top left quadrilateral commutes. First note that the diagram below commutes since the left triangle is by the definition of duals, and the right is clear:

$$\begin{tikzcd}
N \otimes A \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{coev}_N"] \arrow[dr, "\mathrm{id} \otimes \mathrm{id}"'] & N \otimes A \otimes {}^*N \otimes N \arrow[r, "q \otimes \mathrm{id} \otimes \mathrm{id}"] \arrow[d, "\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{ev}_N"] & N \otimes {}^*N \otimes N \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_N"] & N \\
& N \otimes A \arrow[urr, "q"] & &
\end{tikzcd}$$

Then notice that the perimeter of the following diagram is the desired commuting quadrilateral, with the left slide clearly commuting, and the right side commuting by the above tensored with identity:

$$\begin{tikzcd}
A \arrow[r, "\mathrm{coev}_N \otimes \mathrm{id}"] \arrow[d, "\mathrm{id} \otimes \mathrm{coev}_N"] & {}^*N \otimes N \otimes A \arrow[d, "\mathrm{id}^{\otimes 3} \otimes \mathrm{coev}_N"'] \arrow[drr, "\mathrm{id} \otimes q"] & & \\
A \otimes {}^*N \otimes N \arrow[r, "\mathrm{coev}_N \otimes \mathrm{id}^{\otimes 3}"'] & {}^*N \otimes N \otimes A \otimes {}^*N \otimes N \arrow[r, "\mathrm{id} \otimes q \otimes \mathrm{id} \otimes \mathrm{id}"'] & {}^*N \otimes N \otimes {}^*N \otimes N \arrow[r, "\mathrm{id}\, \mathrm{ev}_N \otimes \mathrm{id}"'] & {}^*N \otimes N
\end{tikzcd}$$

Now we wish to construct an inverse $\psi : \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(\mathcal{A})}(M, X \otimes N) \to \mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, X)$. To this end, let $g \in \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(\mathcal{A})}(M, X \otimes N) \subseteq \mathrm{Hom}_{\mathcal{C}}(M, X \otimes N)$, and have g' be its image under $\mathrm{Hom}_{\mathcal{C}}(M, X \otimes N) \cong \mathrm{Hom}_{\mathcal{C}}(M \otimes {}^*N, X)$. I.e. g' is the following:

$$g' : M \otimes {}^*N \xrightarrow{g \otimes \mathrm{id}} X \otimes N \otimes {}^*N \xrightarrow{\mathrm{id} \otimes \mathrm{ev}_N} X$$

We will construct $\psi(g)$ with the universal property of coequalizers as below:

$$\begin{tikzcd}
M \otimes A \otimes N \arrow[r, shift left, "p \otimes \mathrm{id}"] \arrow[r, shift right, "\mathrm{id} \otimes q'"'] & M \otimes {}^*N \arrow[r, "e"] \arrow[dr, "g'"] & M \otimes_A {}^*N \arrow[d, dashed, "\psi(g)"] \\
& & X
\end{tikzcd}$$

However, to do this, we must demonstrate that $g' \circ (p \otimes \mathrm{id}) = g' \circ (\mathrm{id} \otimes q')$. In the diagram below, the bottom left quadrilateral commutes since g is a map of A-modules, and the top quadrilateral clearly commutes. It will be left to check the bottom right one.

$$\begin{tikzcd}
M \otimes A \otimes {}^*N \arrow[rr, "\mathrm{id} \otimes q'"] \arrow[dr, "g \otimes \mathrm{id} \otimes \mathrm{id}"] \arrow[dd, "p \otimes \mathrm{id}"] & & M \otimes {}^*N \arrow[d, "g \otimes \mathrm{id}"] \\
& X \otimes N \otimes A \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes q"] \arrow[d, "\mathrm{id} \otimes q \otimes \mathrm{id}"] & X \otimes N \otimes {}^*N \arrow[d, "\mathrm{id} \otimes \mathrm{ev}_N"] \\
M \otimes {}^*N \arrow[r, "g \otimes \mathrm{id}"] & X \otimes N \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_N"] & X
\end{tikzcd}$$

By removing the extraneous X to the left and expanding q', the perimeter of the following diagram commuting will imply that the above desired square commutes:

$$\begin{tikzcd}
N \otimes A \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{coev}_N \otimes \mathrm{id}"] \arrow[dr, "\mathrm{id}"] & N \otimes {}^*N \otimes N \otimes A \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes q \otimes \mathrm{id}"] \arrow[d, "\mathrm{ev}_N \otimes \mathrm{id}^{\otimes 3}"] & N \otimes {}^*N \otimes N \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{ev}_N"] \arrow[d, "\mathrm{ev}_N \otimes \mathrm{id} \otimes \mathrm{id}"] & N \otimes {}^*N \arrow[d, "\mathrm{ev}_N"] \\
& N \otimes A \otimes {}^*N \arrow[r, "q \otimes \mathrm{id}"] & N \otimes {}^*N \arrow[r, "\mathrm{ev}_N"] & 1
\end{tikzcd}$$

The left triangle commutes since ${}^*N$ is a right dual, and the two squares clearly commute.

Thus we have our map $\psi : \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(\mathcal{A})}(M, X \otimes N) \to \mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, X)$. We must now show that it is the inverse of ϕ. Let $f : M \otimes_A {}^*N \to X$. Notice that $\phi(f)'$ is the image of $f \circ e$ through $\mathrm{Hom}_{\mathcal{C}}(M \otimes {}^*N, X) \cong \mathrm{Hom}_{\mathcal{C}}(M, X \otimes N) \cong \mathrm{Hom}_{\mathcal{C}}(M \otimes {}^*N, X)$, and is hence equal to $f \circ e$ by Theorem 1.17, so we get the following:

$$\begin{tikzcd}
M \otimes A \otimes N \arrow[r, shift left, "p \otimes \mathrm{id}"] \arrow[r, shift right, "\mathrm{id} \otimes q'"'] & M \otimes {}^*N \arrow[r, "e"] \arrow[dr, "\phi(f)'"] & M \otimes_A {}^*N \arrow[d, dashed, "f"] \\
& & X
\end{tikzcd}$$

Thus by the universal property of coequalizers, $\psi(\phi(f)) = f$. Now, for the other direction, let $g : M \to X \otimes N$. Then by definition we have

$$\begin{tikzcd}
M \otimes A \otimes N \arrow[r, shift left, "p \otimes \mathrm{id}"] \arrow[r, shift right, "\mathrm{id} \otimes q'"'] & M \otimes {}^*N \arrow[r, "g \otimes \mathrm{id}"] \arrow[dr, "e"'] & X \otimes N \otimes {}^*N \arrow[r, "\mathrm{id} \otimes \mathrm{ev}_N"] & X \\
& & M \otimes_A {}^*N \arrow[ur, "\psi(g)"'] &
\end{tikzcd}$$

So then the top path of the below diagram is $\phi(\psi(g))$, the bottom is g by the definition of duals, the left square clearly commutes, and the right is by the above diagram.

$$\begin{tikzcd}
M \arrow[r, "\mathrm{id} \otimes \mathrm{coev}_N"] \arrow[dr, "g"'] & M \otimes {}^*N \otimes N \arrow[r, "e \otimes \mathrm{id}"] \arrow[dr, "g \otimes \mathrm{id}"'] & (M \otimes_A {}^*N) \otimes N \arrow[r, "\psi(g) \otimes \mathrm{id}"] & X \otimes N \\
& X \otimes N \arrow[r, "\mathrm{id} \otimes \mathrm{id} \otimes \mathrm{coev}_N"] & X \otimes N \otimes {}^*N \otimes N \arrow[ur, "\mathrm{id} \otimes \mathrm{ev}_N"'] &
\end{tikzcd}$$

Thus we have that ϕ and ψ are mutually inverse, showing that we have an isomorphism:

$$\mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, X) \cong \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(A)}(M, X \otimes N)$$

Now it remains to see naturality. Recall that given $h : (M', p') \to (M, p)$, a map of right A-modules, and (B, r) a left A-module, we get a map $h \otimes_A \mathrm{id} : M' \otimes_A B \to M \otimes_A B$ defined as below:

$$\begin{tikzcd}
M' \otimes A \otimes B \arrow[r, shift left, "p' \otimes \mathrm{id}"] \arrow[r, shift right, "\mathrm{id} \otimes r"'] & M' \otimes B \arrow[r, "e'"] \arrow[d, "h \otimes \mathrm{id}"] & M' \otimes_A B \arrow[d, dashed, "h \otimes_A \mathrm{id}"] \\
& M \otimes B \arrow[r, "e"] & M \otimes_A B
\end{tikzcd}$$

Where e' is the equalizer map for $M' \otimes_A {}^*N$ and the requirements for invoking the universal property of the coequalizer follows readily from the fact that f is a morphism of modules. Let $h : M' \to M$ be as above, $k : X \to X'$ be a map in $\mathcal{C}$, and $f \in \mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, X)$. In the diagram below, the top path is $[\phi \circ \mathrm{Hom}(h \otimes_A \mathrm{id}, k)]$, and the bottom path is $[\mathrm{Hom}(h, k \otimes \mathrm{id}) \circ \phi]$.

$$\begin{tikzcd}
M' \arrow[r, "\mathrm{id} \otimes \mathrm{coev}_N"] \arrow[d, "h"] & M' \otimes {}^*N \otimes N \arrow[r, "e' \otimes \mathrm{id}"] \arrow[d, "h \otimes \mathrm{id} \otimes \mathrm{id}"] & (M' \otimes_A {}^*N) \otimes N \arrow[r, "(h \otimes_A \mathrm{id}) \otimes \mathrm{id}"] & (M \otimes_A {}^*N) \otimes N \arrow[r, "f \otimes \mathrm{id}"] \arrow[dl, equal] \arrow[d, "f \otimes \mathrm{id}"] & X \otimes N \arrow[d, "k \otimes \mathrm{id}"] \\
M \arrow[r, "\mathrm{id} \otimes \mathrm{coev}_N"] & M \otimes {}^*N \otimes N \arrow[r, "e \otimes \mathrm{id}"] & (M \otimes_A {}^*N) \otimes N \arrow[r, "f \otimes \mathrm{id}"] & X \otimes N \arrow[r, "k \otimes \mathrm{id}"] & X' \otimes N
\end{tikzcd}$$

The left and right squares commute, the center polygon commutes by definition of $h \otimes_A \mathrm{id}$, and the triangle trivially commutes. Thus the isomorphism is natural, and we have an adjunction. $\square$

Corollary 3.11. Let A be an algebra in $\mathcal{C}$, a fusion category. Let (M, p) and (N, q) be right A-modules. Then $\underline{\mathrm{Hom}}(M, N) \cong (M \otimes_A {}^*N)^*$.

Proof. This is witnessed by the following chain of natural isomorphisms:

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(A)}(X \otimes M, N) &\cong \mathrm{Hom}_{\mathrm{Mod}_{\mathcal{C}}(A)}(M, {}^*X \otimes N) \\
&\cong \mathrm{Hom}_{\mathcal{C}}(M \otimes_A {}^*N, {}^*X) \\
&\cong \mathrm{Hom}_{\mathcal{C}}(({}^*X)^*, (M \otimes_A {}^*N)^*) \\
&\cong \mathrm{Hom}_{\mathcal{C}}(X, (M \otimes_A {}^*N)^*)
\end{aligned}$$

The second isomorphism is the above theorem, and the last is because $({}^*X)^* \cong X$. $\square$

3.2 Finite Dimensional (Linear) Representations of Finite Groups

Our first motivating example of what is to come is going to be the tensor products of finite dimensional representations over a finite group. Recall that $\mathrm{Rep}(G) = \mathrm{Vec}^G$ for a finite group G. We have the functor which takes a pair of representations to their tensor product:

$$\otimes : \mathrm{Rep}(G) \times \mathrm{Rep}(G) \to \mathrm{Rep}(G); (V, \rho) \otimes (W, \theta) := (V \otimes W, \rho \otimes \theta)$$

Where $[\rho \otimes \theta]_g(v \otimes w) = \rho_g(V) \otimes \theta_g(w)$. This exhibits $\mathrm{Rep}(G)$ as a $\mathrm{Rep}(G)$-module category. Thus we may ask what its internal homs are.

Since there is a forgetful functor $\mathrm{Rep}(G) \to \mathrm{Vec}$, one would guess that the underlying object of the internal hom in $\mathrm{Rep}(G)$ may be $\mathrm{Hom}_{\mathrm{Vec}}(-,-)$. Indeed one can form a canonical representation $\mathrm{Hom}_{\mathrm{Vec}}(V, W)$ as follows:

$$\mathrm{Hom}(\rho, \theta)_g : \mathrm{Hom}_{\mathrm{Vec}}(V, W) \to \mathrm{Hom}_{\mathrm{Vec}}(V, W); f(v) \mapsto \theta_g(f(\rho_g^{-1}(v)))$$

This can also be derived by translating the action on $W \otimes V^*$ to $\mathrm{Hom}_{\mathrm{Vec}}(V, W)$ with the isomorphism from Example 1.19.

Theorem 3.12. *There is an adjunction as follows:*

$$\mathrm{Hom}_{\mathrm{Rep}(G)}((U, \pi) \otimes (V, \rho), (W, \theta)) \cong \mathrm{Hom}_{\mathrm{Rep}(G)}((U, \pi), (\mathrm{Hom}_{\mathrm{Vec}}(V, W), \mathrm{Hom}(\rho, \theta)))$$

In other words, $\mathrm{Hom}_{\mathrm{Vec}}(V, W)$ *equipped with the action* $\mathrm{Hom}(\rho, \theta)$ *is an internal hom.*

Proof. Follows from the fact that $W \otimes V^* \cong \mathrm{Hom}_{\mathrm{Vec}}(V, W)$ (Example 1.19) and Theorem 1.17. ☐

If $H \leq G$, then we have the following *restriction functor* of representations:

$$\mathrm{Res} : \mathrm{Rep}(G) \to \mathrm{Rep}(H); (V, \rho) \mapsto (\mathrm{Res}(V) := V, \mathrm{Res}(\rho) := \rho|_H)$$

For a morphism $f : V \to W$, $\mathrm{Res}(f) = f$, so it is trivial on morphisms. This is clearly well defined, since the required diagrams still commute.

This allows us to restrict representations from G to H. There is also an induction functor, allowing us to get representations of G from representations of H. We define this functor below:

Choose a system of representatives $\{g_i\}_{i\in I}$ for the cosets G/H. I.e. $G/H = \{g_iH\}_{i\in I}$. For $g \in G$, and $i \in I$ we have that $gg_i = g_{j(g,i)}h_{g,i}$ with $h_{g,i} \in H$. Let $(V, \rho) \in \mathrm{Rep}(H)$, for $i \in I$, put $V_i = V$ and define $\rho^i_g : V_i \to V_{j(g,i)}$ as $\rho_{h_{g,i}}$. Recall that action by left multiplication by g on G/H is a permutation, so $j(g, -) : I \to I$ is a bijection. Then we can define *induction* as follows:

$$\mathrm{Ind} : \mathrm{Rep}(H) \to \mathrm{Rep}(G); (V, \rho) \mapsto (\mathrm{Ind}(V) := \bigoplus_i V_i, \mathrm{Ind}(\rho) := \bigoplus_i \rho^i)$$

Unpacking this a bit, we have the following action:

$$\mathrm{Ind}(\rho)_g((v_i)_{i\in I}) = (\rho_{h_{g,i}}(v_i))_{j(g,i)\in I}$$

On morphisms $f : V \to W$, $\mathrm{Ind}(f) = \bigoplus_i f$ is just a direct sum of copies $f : V_i \to W_i$. Since $\mathrm{Ind}(f)$ is just f on all components, the resulting map will be a morphism.

Remark 3.13. If one wants to specify which pair $H \leq G$ one is restricting/inducting on, one can decorate the functors as Res^G_H and Ind^G_H accordingly.

We will now show that this is well defined. If we have $k, g \in G$, then we have:

$$\begin{aligned}
(kg)g_i =& k(gg_i) \\
=& k(g_{j(g,i))}h_{g,i}) \\
=& (kg_{j(g,i))})h_{g,i} \\
=& (g_{j(k,j(g,i))}h_{k,j(g,i))})h_{g,i} \\
=& g_{j(k,j(g,i))}(h_{k,j(g,i)}h_{g,i})
\end{aligned}$$

So $j(kg, i) = j(k, j(g, i))$, and $h_{kg,i} = h_{k,j(g,i)}h_{g,i}$. Then we compute the composition below:

$$\begin{aligned}
[\mathrm{Ind}(\rho)_k \circ \mathrm{Ind}(\rho)_g]((v_i)_i) =& \mathrm{Ind}(\rho)_k[\mathrm{Ind}(\rho)_g((v_i)_i)] \\
=& \mathrm{Ind}(\rho)_k[(\rho_{h_{g,i}}(v_i))_{j(g,i)}] \\
=& (\rho_{h_{k,j(g,i))}}(\rho_{h_{g,i}}(v_i)))_{j(k,j(g,i))} \\
=& (\rho_{h_{k,j(g,i))}h_{g,i}}(v_i))_{j(k,j(g,i))} \\
=& (\rho_{h_{kg},i}(v_i))_{j(k,j(g,i))} \\
=& \mathrm{Ind}(\rho)_{kg}((v_i)_i)
\end{aligned}$$

Thus $\mathrm{Ind}(V) \in \mathrm{Rep}(G)$, and our functor Ind is well defined.

Remark 3.14. There are easier ways to define this functor, but we choose this particular presentation so that the proof of Theorem 3.15 will generalize to the proof of Theorem 3.22.

Theorem 3.15 (Frobenius Reciprocity)**.** *Let $(V, \rho) \in \mathrm{Rep}(G)$ and $(W, \theta) \in \mathrm{Rep}(H)$ with $H \leq G$ and G a finite group. Then we have the following adjunction:*

$$\mathrm{Hom}_{\mathrm{Rep}(H)}(\mathrm{Res}(V), W) \cong \mathrm{Hom}_{\mathrm{Rep}(G)}(V, \mathrm{Ind}(W))$$

Proof. Decompose 1_G as $g_t l$ with $l \in H$. Then we have the following "evaluation map":

$$e : \mathrm{Ind}(W) \to W; (w_i)_i = \theta_{g_t}(v_t)$$

We claim that this is an H-equivarient map. First notice that since $1_g \in H$, and $l \in H$, that $g_t \in H$. Thus if $h \in H$ and i is such that $j(h, i) = t$, we have that $hg_i = g_t h_{h,i}$. Since $h, g_t, h_{h,i} \in H$, so is g_i. Hence $i = t$. Then we get that $hg_t = g_t h_{h,t}$, which one can easily show implies that $h_{h,t} = lhg_t$. Now, we can compute:

$$\begin{aligned}
e(\mathrm{Ind}(\theta)_h((w_i)_i)) =& e((\theta_{h_{h,i}}(w_i))_{j(h,i)}) \\
=& \theta_{g_t}(\theta_{lhg_t}(w_t)) \\
=& \theta_{hg_t}(w_t) \\
=& \theta_h(\theta_{g_t}(w_t)) \\
=& \theta_h(e((w_i)_i))
\end{aligned}$$

Now, given $p : V \to \mathrm{Ind}(W)$, a map of G-reps, we have $\psi(p) := e \circ p : \mathrm{Res}(V) \to W$, which is a map of H-reps since it is a composition of maps of H-reps. Thus we have a map:

$$\psi : \mathrm{Hom}_{\mathrm{Rep}(G)}(V, \mathrm{Ind}(W)) \to \mathrm{Hom}_{\mathrm{Rep}(H)}(\mathrm{Res}(V), W)$$

In the other direction, given $q : \mathrm{Res}(V) \to W$, a map of H-reps, let $\varphi(q) : \mathrm{V} \to \mathrm{Ind}(W)$ be given as below:

$$\varphi(q)(v) = (q(\rho_{g_i^{-1}}(v)))_i$$

To show that this is a map of G-reps, we compute:

$$\begin{aligned}
\mathrm{Ind}(\theta)_g(\varphi(q)(v)) =& \mathrm{Ind}(\theta)_g((q(\rho_{g_i^{-1}}(v)))_i) \\
=& (\theta_{h_{g,i}}(q(\rho_{g_i^{-1}}(v))))_{j(g,i)} \\
=& (q(\rho_{h_{g,i}}(\rho_{g_i^{-1}}(v))))_{j(g,i)} \\
=& (q(\rho_{h_{g,i}g_i^{-1}}(v)))_{j(g,i)} \\
=& (q(\rho_{g_\ell^{-1}g}(v)))_\ell \qquad (*) \\
=& (q(\rho_{g_\ell^{-1}}(\rho_g(v)))_\ell \\
=& \varphi(q)(\rho_g(v))
\end{aligned}$$

Note that on the line with the $(*)$ we reindexed to $\ell = j(g,i)$ so that we would have that $g_\ell^{-1}g = h_{g,i}g_i^{-1}$, which can easily be derived from $gg_i = g_\ell h_{g,i}$. So we get another map:

$$\varphi : \mathrm{Hom}_{\mathrm{Rep}(H)}(\mathrm{Res}(V), W) \to \mathrm{Hom}_{\mathrm{Rep}(G)}(V, \mathrm{Ind}(W))$$

To see that ψ and φ are mutually inverse, we have the following computations:

$$\begin{aligned}
\psi(\varphi(q))(v) =& \psi((q(\rho_{g_i^{-1}}(-)))_i)(v) \\
=& e((q(\rho_{g_i^{-1}}(v)))_i) \\
=& \theta_{g_t}(q(\rho_{g_t^{-1}}(v))) \\
=& q(\rho_{g_t}(\rho_{g_t^{-1}}(v))) \\
=& q(\rho_{g_t(g_t)^{-1}}(v)) \\
=& q(\rho_{1_g}(v)) \\
=& q(v)
\end{aligned}$$

$$\begin{aligned}
\varphi(\psi(p))(v) =& \varphi(e \circ p)(v) \\
=& (e(p(\rho_{g_i^{-1}}(v))))_i \\
=& (e(\mathrm{Ind}(\theta)_{g_i^{-1}}(p(v))))_i \\
=& (e(\mathrm{Ind}(\theta)_{g_i^{-1}}((p(v)_j)_j)))_i \\
=& e((\theta_{h_{g_i^{-1},j}}(p(v)_j)))_{j(g_i^{-1},j)})_i \\
=& (\theta_{g_t}(\theta_l(p(v)_i)))_i \qquad (*) \\
=& (p(v)_i)_i = p(v)
\end{aligned}$$

Note that on the starred line, in order to get the indices to match up, we set $t = j(g_i^{-1}, j)$ and solved for j. We have that $g_i^{-1}g_j = g_t h_{g_i^{-1},j}$, but the right side is in H, so g_i^{-1} and g_j must be in the same coset. Thus $i = j$, and we get that $1_G = g_t h_{g_i^{-1},j}$, so $h_{g_i^{-1},j} = l$. Thus ψ and φ are mutually inverse.

For naturality, let $\alpha : V' \to V$ and $\beta : W \to W'$ be maps of G and H representations respectively. Let $p : V \to \mathrm{Ind}(W)$ be a map of G-representations, $v \in V'$ and compute:

$$\begin{aligned}
[\mathrm{Hom}(\mathrm{Res}(\alpha), \beta) \circ \psi](p)(v) =& \beta(\psi(p)(\alpha(v))) \\
=& \beta(e(p(\alpha(v)))) \\
=& \beta(e((p(\alpha(v)_i))_i)) \\
=& \beta(\theta_{g_t}(p(\alpha(v))_t)) \\
=& \theta'_{g_t}(\beta(p(\alpha(v)_t))) \\
=& e((\beta(p(\alpha(v))_i)_i) \\
=& e([\mathrm{Hom}(\alpha, \mathrm{Ind}(\beta))(p)](v)) \\
=& \psi(\mathrm{Hom}(\alpha, \mathrm{Ind}(\beta)))(p)(v)
\end{aligned}$$

Thus the isomorphism is natural, and hence we get our desired adjunction. □

Remark 3.16. In this case we have an adjunction $\mathrm{Res} \dashv \mathrm{Ind}$, but in general for infinite groups, we have $\mathrm{Ind} \dashv \mathrm{Res} \dashv \mathrm{Coind}$. It turns out that in the finite case, Ind and Coind are naturally isomorphic and we have $\mathrm{Ind} \dashv \mathrm{Res} \dashv \mathrm{Ind}$ (see Lorenz (2018) Section 3.1.4.).

With the restriction functor, we can demonstrate $\mathrm{Rep}(H)$ as a $\mathrm{Rep}(G)$-module category:

$$\otimes^H : \mathrm{Rep}(G) \times \mathrm{Rep}(H) \to \mathrm{Rep}(H); (V, \rho) \otimes^H (W, \theta) := \mathrm{Res}(V) \otimes W = (V \otimes W, \rho|_H \otimes \theta)$$

Corollary 3.17. There is an adjunction:

$$\mathrm{Hom}_{\mathrm{Rep}(H)}((U, \pi) \otimes^H (V, \rho), (U, \theta)) \cong \mathrm{Hom}_{\mathrm{Rep}(G)}((U, \pi), \mathrm{Ind}(\mathrm{Hom}_{\mathrm{Vec}}(V, W)))$$

In other words, $\mathrm{Ind}(\mathrm{Hom}_{\mathrm{Vec}}(V, W))$ is the internal hom of $\mathrm{Rep}(H)$ as a $\mathrm{Rep}(G)$-module.

Proof. We have the following composition of natural isomorphisms:

$$\begin{aligned}
\mathrm{Hom}_{\mathrm{Rep}(H)}((U, \pi) \otimes^H (V, \rho), (W, \theta)) &= \mathrm{Hom}_{\mathrm{Rep}(H)}(\mathrm{Res}(U, \pi) \otimes (V, \rho), (W, \theta)) \\
&\cong \mathrm{Hom}_{\mathrm{Rep}(H)}(\mathrm{Res}(U, \pi), \mathrm{Hom}_{\mathrm{Vec}}(V, W)) \\
&\cong \mathrm{Hom}_{\mathrm{Rep}(G)}((U, \pi), \mathrm{Ind}(\mathrm{Hom}_{\mathrm{Vec}}(V, W))
\end{aligned}$$

Where the first is Theorem 3.12 applied to H, and the second is Theorem 3.15. □

3.3 Finite Dimensional (Projective) Representations of Finite Groups

Recall that given a 2-cocycle $\mu \in Z^2(H, \Bbbk)$, we can form the category $\operatorname{Rep}_\mu(H)$ of projective representations with Schur multiplier μ. $\operatorname{Rep}_\mu(H)$ is a $\operatorname{Rep}(G)$-module category with the action

$$\otimes^{H,\mu} : \operatorname{Rep}(G) \times \operatorname{Rep}_\mu(H) \to \operatorname{Rep}_\mu(H); (V, \rho) \otimes^{H,\mu} (W, \theta) := \operatorname{Res}(V) \otimes W$$

This is well defined since $\mathrm{id}_V \otimes(\mu(k, h)\,\mathrm{id}_W) = \mu(k, h)(\mathrm{id}_{V\otimes W})$, so the following commutes:

$$\begin{tikzcd}
V \otimes W \arrow[r, "\rho_h\otimes\theta_h"] \arrow[d, "\mu(k,h)\,\mathrm{id}_{V\otimes W}"] & V \otimes W \arrow[d, "\rho_k\otimes\theta_k"] \\
V \otimes W \arrow[r, "\rho_{kh}\otimes\theta_{kh}"] & V \otimes W
\end{tikzcd}$$

Lemma 3.18. Given $(V, \rho), (W, \theta) \in \operatorname{Rep}_\mu(H)$, then $(\operatorname{Hom}_{\mathrm{Vec}}(V, W), \operatorname{Hom}(\rho, \theta)) \in \operatorname{Rep}(H)$.

Proof. To see this we have the following computation:

$$\begin{aligned}
[\operatorname{Hom}(\rho, \theta)_k \circ \operatorname{Hom}(\rho, \theta)_h](f) =& \operatorname{Hom}(\rho, \theta)_k[\theta_h(f(\rho_h^{-1}(-)))] \\
=&\theta_k(\theta_h(f(\rho_h^{-1}(\rho_k^{-1}(-))))) \\
=&[\theta_k \circ \theta_h](f([\rho_h^{-1} \circ \rho_k^{-1}](-))) \\
=&[\mu(k, h)\theta_{kh}](f([(\rho_k \circ \rho_h)^{-1}](-))) \\
=&\mu(k, h)\theta_{kh}(f(\mu(k, k)^{-1}\rho_{kh}^{-1})) \\
=&\mu(k, h)\mu(k, h)^{-1}\theta_{kh}(f(\rho_{kh}^{-1}(-))) \\
=& \operatorname{Hom}(\rho, \theta)_{kh}(f)
\end{aligned}$$

□

Here we really see why considering internal homs with respect to a module category rather than the category itself makes sense. The hom of projective representations over μ is just a normal representation, so it wouldn't stay internal to the category $\operatorname{Rep}_\mu(H)$. In this sense, they aren't really "internal" anymore, but the vocabulary is borrowed from the more well known notion introduced at the beginning of chapter 3.

Theorem 3.19. *Let μ be as above, then we have an adjunction:*

$$\mathrm{Hom}_{\mathrm{Rep}_\mu(H)}((U,\pi)\otimes^{H,\mu}(V,\rho),(W,\theta))\cong \mathrm{Hom}_{\mathrm{Rep}(G)}((U,\pi),\mathrm{Ind}(\mathrm{Hom}_{\mathrm{Vec}}(V,W)))$$

In other words, $\mathrm{Ind}(\mathrm{Hom}_{\mathrm{Vec}}(V,W))$ *is the internal hom of* $\mathrm{Rep}_\mu(H)$ *as a* $\mathrm{Rep}(G)$*-module.*

Proof. Let us first consider the case when $H=G$. Then Res and Ind are both trivial, so it amounts to showing the following:

$$\mathrm{Hom}_{\mathrm{Rep}_\mu(H)}((U,\pi)\otimes^{H,\mu}(V,\rho),(W,\theta))\cong \mathrm{Hom}_{\mathrm{Rep}(H)}((U,\pi),(\mathrm{Hom}_{\mathrm{Vec}}(V,W),\mathrm{Hom}(\rho,\theta)))$$

Since a homomorphism of projective representations is defined exactly the same as a homomorphism of non-projective representations, proving this is identical to the proof of Theorem 3.12, which boils down to the proof of Theorem 1.17. So then we have that:

$$\begin{aligned}\mathrm{Hom}_{\mathrm{Rep}_\mu(H)}((U,\pi)\otimes^{H,\mu}(V,\rho),(W,\theta)) &= \mathrm{Hom}_{\mathrm{Rep}_\mu(H)}(\mathrm{Res}(U,\pi)\otimes(V,\rho),(W,\theta))\\ &\cong \mathrm{Hom}_{\mathrm{Rep}(H)}(\mathrm{Res}(U,\pi),\mathrm{Hom}_{\mathrm{Vec}}(V,W))\\ &\cong \mathrm{Hom}_{\mathrm{Rep}(G)}((U,\pi),\mathrm{Ind}(\mathrm{Hom}_{\mathrm{Vec}}(V,W))\end{aligned}$$

Where now $(U,\pi)\in\mathrm{Rep}(G)$, and we applied the first case to $\mathrm{Res}(U,\pi)$. □

Remark 3.20. There is some notational ambiguity going on here, since the action is denoted $\otimes^{H,\mu}$, but that only specifies the module category, not which category it's a module over. So in the right side of the first equality, we write $\mathrm{Res}(U,\pi)\otimes(V,\rho)$ to mean the action of $\mathrm{Rep}(H)$ on $\mathrm{Rep}_\mu(H)$.

This result agrees with the discussion above Lemma 4.3 in Etingof and Ostrik (2003), which stated that for a simple $V\in\mathrm{Rep}_\mu(H)$, $\underline{\mathrm{Hom}}(V,V)=\mathrm{Ind}(\mathrm{End}(V))$.

3.4 Representations of Finite Groups in a Fusion Category

Notice that our analysis for the internal homs in section 3.2 was made much simpler by the fact that the action of G on Vec was trivial, i.e. that actions gave automorphisms $X\xrightarrow{\sim}X$. More generally, if G acts trivially on a fusion category $\mathcal{C}$, this yeilds that objects $(X,u)\in\mathcal{C}^G$ look like a more modest generalizations of categories of finite dimensional representations of G since $u_g : X \xrightarrow{\sim} X$. Indeed, these are categories of representations of G valued in the fusion category $\mathcal{C}$, as we define below:

Definition 3.21. For a fusion category $\mathcal{C}$, define the category of *representations of G in $\mathcal{C}$*, by $\mathrm{Rep}_{\mathcal{C}}(G) := \mathcal{C}^G$ with respect to the trivial action of a finite group G on $\mathcal{C}$. Explicitly, objects of this category are pairs of an object $X \in \mathcal{C}$, and a family of automorphisms $u = \{u_g\}_{g\in G}$ of X such that $u_g \circ u_h = u_{gh}$, and the morphisms are intertwining operators. Clearly we have that $\mathrm{Rep}_{\mathrm{Vec}}(G) = \mathrm{Rep}(G)$.

For $H \leq G$, there is an obvious restriction functor:

$$\mathrm{Res} : \mathrm{Rep}_{\mathcal{C}}(G) \to \mathrm{Rep}_{\mathcal{C}}(H); (X, \{u_g\}_{g\in G}]) \mapsto (X, \{u_h\}_{h\in H})$$

This demonstrates $\mathrm{Rep}_{\mathcal{C}}(H)$ as a $\mathrm{Rep}_{\mathcal{C}}(G)$-module category as follows:

$$\otimes^H : \mathrm{Rep}_{\mathcal{C}}(G) \times \mathrm{Rep}_{\mathcal{C}}(H) \to \mathrm{Rep}_{\mathcal{C}}(H); ((X,u),(Y,v)) \mapsto (\mathrm{Res}(X)\otimes Y, \mathrm{Res}(u)\otimes v)$$

We wish to generalize the induction functor from the case of $\mathrm{Rep}(G)$. Let $H \leq G$ and again choose representatives $\{g_i\}_{i\in I}$ for the cosets G/H and define $gg_i = g_{j(g,i)}h_{g,i}$, $X_i := X$, and $u_g^i : X_i \to X_{j(g,i))}$ by $u_{h_{g,i}}$. Then we have:

$$\mathrm{Ind} : \mathrm{Rep}_{\mathcal{C}}(H) \to \mathrm{Rep}(G); (X, \{u_h\}_{h\in H}) \mapsto (\mathrm{Ind}(X) := \bigoplus_i X_i, \{\mathrm{Ind}(u)_g := \bigoplus_i u_g^i\}_{g\in G})$$

Let $\mathrm{pr}_i : \mathrm{Ind}(X) \to X$ be projection onto the ith component given by the definition of direct sums. More specifically, to define the map $\mathrm{Ind}(u)_g$ we use the universal property of the product. In other words, for each $\ell \in I$ we require that the following commutes:

$$\begin{tikzcd}
\mathrm{Ind}(X) = \bigoplus_i X_i \arrow[r, "\mathrm{pr}_\ell"] \arrow[d, dashed, "\mathrm{Ind}(u)_g"] & X_\ell \arrow[d, "u_g^\ell"] \\
\mathrm{Ind}(X) = \bigoplus_i X_i \arrow[r, "\mathrm{pr}_{j(g,\ell))}"] & X_{j(g,\ell))}
\end{tikzcd}$$

Let us see that this is well defined on objects. Let $k, g \in G$, and $\ell \in I$, and refer to the following:

$$\begin{tikzcd}
\mathrm{Ind}(X) \arrow[r, "\mathrm{Ind}(u)_g"] \arrow[d, "\mathrm{pr}_\ell"] & \mathrm{Ind}(X) \arrow[r, "\mathrm{Ind}(u)_k"] \arrow[d, "\mathrm{pr}_{j(g,\ell)}"] & \mathrm{Ind}(X) \arrow[d, "\mathrm{pr}_{j(k,j(g,\ell))}"] \\
X_\ell \arrow[r, "u_g^\ell"] & X_{j(g,\ell)} \arrow[r, "u_k^{j(g,\ell)}"] & X_{j(k,j(g,\ell))}
\end{tikzcd}$$

Each square commuting is by definition of $\mathrm{Ind}(u)_{-}$. Notice that the bottom path is $u_{kg}^{j(g,\ell)}$, and $j(kg,\ell) = j(k, j(g,\ell))$. Hence by the uniqueness in the universal property of the product, the top path is equal to $\mathrm{Ind}(u)_{kg}$. Thus $\mathrm{Ind}(u)_k \circ \mathrm{Ind}(u)_g = \mathrm{Ind}(u)_{kg}$.

Given a map $f : (X,u) \to (Y,v)$ in $\mathrm{Rep}_{\mathcal{C}}(H)$, define $\mathrm{Ind}(f)$ as $\bigoplus_i f$, or more specifically by universal property of the product as below:

$$\begin{tikzcd}
\mathrm{Ind}(X) \arrow[r, "\mathrm{pr}_\ell"] \arrow[d, dashed, "\mathrm{Ind}(f)"] & X_\ell \arrow[d, "f"] \\
\mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_\ell"] & Y_\ell
\end{tikzcd}$$

To see that $\mathrm{Ind}(f) \in \mathrm{Rep}_{\mathcal{C}}(G)$, let $g \in G$, $\ell \in I$ and refer to the below diagram:

$$\begin{tikzcd}
\mathrm{Ind}(X) \arrow[rrr, "\mathrm{Ind}(f)"] \arrow[dr, "\mathrm{pr}_\ell"] \arrow[ddd, "\mathrm{Ind}(u)_g"] & & & \mathrm{Ind}(Y) \arrow[dl, "\mathrm{pr}_\ell"'] \arrow[ddd, "\mathrm{Ind}(v)_g"] \\
& X_\ell \arrow[r, "f"] \arrow[d, "u_g^\ell"] & Y_\ell \arrow[d, "v_g^\ell"] & \\
& X_{j(g,\ell)} \arrow[r, "f"] & Y_{j(g,\ell)} & \\
\mathrm{Ind}(X) \arrow[rrr, "\mathrm{Ind}(f)"] \arrow[ur, "\mathrm{pr}_{j(g,\ell)}"] & & & \mathrm{Ind}(Y) \arrow[ul, "\mathrm{pr}_{j(g,\ell)}"']
\end{tikzcd}$$

The top and bottom quadrilaterals commute by definition of $\mathrm{Ind}(f)$, the left and right quadrilaterals commute by definition of $\mathrm{Ind}(u)$ and $\mathrm{Ind}(v)$, and the middle commutes since $f \in \mathrm{Rep}_{\mathcal{C}}(H)$ applied to $h_{g,\ell}$. Since this is for all $\ell \in I$, the uniqueness in the universal property of the product gives that the top and bottom paths are equal, and hence $\mathrm{Ind}(f) \in \mathrm{Rep}_{\mathcal{C}}(G)$.

Theorem 3.22 ($\mathcal{C}$ Frobenius Reciprocity)**.** *Let $(X,u) \in \mathrm{Rep}_{\mathcal{C}}(G)$ and $(Y,v) \in \mathrm{Rep}_{\mathcal{C}}(H)$ with $H \leq G$ and G a finite group. Then we have the following adjunction:*

$$\mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(H)}(\mathrm{Res}(X), Y) \cong \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(G)}(X, \mathrm{Ind}(Y))$$

Proof. Again, decompose 1_G as $g_t l$ with $l \in H$. Then we have the following "evaluation map":

$$e := v_{g_t} \circ \mathrm{pr}_t : \mathrm{Ind}(Y) \to Y$$

This is a map in $\mathcal{C}$ since $g_t \in H$, but we claim that its H-equivariant, i.e. for all $h \in H$, the following commutes:

$$\begin{tikzcd} \mathrm{Ind}(Y) \arrow[r, "e"] \arrow[d, "\mathrm{Ind}(v)_h"] & Y \arrow[d, "v_h"] \\ \mathrm{Ind}(Y) \arrow[r, "e"] & Y \end{tikzcd}$$

Recall from the proof of Theorem 3.15 we have that $j(h,t) = t$, and $h_{h,t} = lhl^{-1} = lhg_t$. Thus the left square on the below diagram commutes by definition of $\mathrm{Ind}(v)_h$:

$$\begin{tikzcd} \mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_t"] \arrow[d, "\mathrm{Ind}(v)_h"] & Y_t = Y \arrow[r, "v_{g_t}"] \arrow[d, "v_{lhg_t}"] & Y \arrow[d, "v_h"] \\ \mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_t"] & Y_t = Y \arrow[r, "v_{gt}"] & Y \end{tikzcd}$$

The right square commutes since both paths are equal to v_{hg_t}, hence the perimeter commutes. The top and bottom sides of the rectangle are e, and so the commutativity of the perimeter proves the claim.

Now we define a map $\psi : \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(G)}(X, \mathrm{Ind}(Y)) \to \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(H)}(\mathrm{Res}(X), Y)$ by sending $p : X \to \mathrm{Ind}(Y)$ to $\psi(p) = e \circ p : \mathrm{Res}(X) = X \to \mathrm{Ind}(Y)$. Since both p and e are H-equivariant, their composition will be as well. Thus it is a morphism in $\mathrm{Rep}_{\mathcal{C}}(H)$, so ψ is well defined.

Conversely, we will define a map $\varphi : \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(H)}(\mathrm{Res}(X), Y) \to \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(G)}(X, \mathrm{Ind}(Y))$ where given $q : \mathrm{Res}(X) \to Y$, we construct $\varphi(q) : V \to \mathrm{Ind}(Y)$ by using the universal property of the product as below:

$$\begin{tikzcd} X \arrow[r, "u_{g_i^{-1}} = u_{g_i}^{-1}"] \arrow[d, dashed, "\varphi(q)"] & X = \mathrm{Res}(X) \arrow[d, "q"] \\ \mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_i"] & Y_i = Y \end{tikzcd}$$

To see that $\varphi(q) \in \mathrm{Rep}_{\mathcal{C}}(G)$, let $\ell \in I$ and see the below diagram:

$$\begin{tikzcd}
X \arrow[rrr, "\varphi(q)"] \arrow[dr, "u_{g_\ell^{-1}}"] \arrow[ddd, "u_g"'] & & & \mathrm{Ind}(Y) \arrow[dl, "\mathrm{pr}_\ell"] \arrow[ddd, "\mathrm{Ind}(v)_g"] \\
& X = \mathrm{Res}(X) \arrow[r, "q"] \arrow[d, "u_{h_{g,\ell}}"] & Y = Y_\ell \arrow[d, "v_{h_{g,\ell}} = v_g^\ell"] & \\
& X = \mathrm{Res}(X) \arrow[r, "q"] & Y = Y_{j(g,\ell)} & \\
X \arrow[ur, "u_{g^{-1}_{j(g_\ell)}}"] \arrow[rrr, "\varphi(q)"] & & & \mathrm{Ind}(Y) \arrow[ul, "\mathrm{pr}_{j(g,\ell)}"']
\end{tikzcd}$$

The top and bottom quadrilaterals commute by definition of $\varphi(q)$. The right quadrilateral commutes by definition of $\mathrm{Ind}(v)_g$, and the middle commutes since q is in $\mathrm{Rep}_{\mathcal{C}}(H)$. Rearranging $gg_\ell = g_{j(g,\ell)}h_{g,\ell}$ we get that $g^{-1}_{j(g,\ell)}g = h_{g,\ell}g_\ell^{-1}$. Thus the left quadrilateral commutes. Since this is for all $\ell \in I$, the uniqueness in the universal property for products gives us that the top path is equal to the bottom. Hence $\varphi(q)$ is a map in $\mathrm{Rep}_{\mathcal{C}}(G)$. Thus it's well defined.

We will now show that ψ and φ are mutually inverse. Let $q : \mathrm{Res}(X) \to Y$. Then we have:

$$\begin{tikzcd}
& & & Y & \\
& \mathrm{Ind}(Y) \arrow[urr, "e"] \arrow[rr, "\mathrm{pr}_t"] & & Y_t = Y \arrow[u, "v_{g_t}"] & \\
X = \mathrm{Res}(X) \arrow[uurrr, bend left, "\psi(\varphi(q))"] \arrow[ur, "\varphi(q)"'] \arrow[rr, "u_{g_t}^{-1}"] & & X = \mathrm{Res}(X) \arrow[ur, "q"] \arrow[rr, "u_{g_t}"] & & X = \mathrm{Res}(X) \arrow[uul, bend right, "q"']
\end{tikzcd}$$

The top left triangle commutes by definition of ψ. The more central triangle commutes by definition of e. The center quadrilateral commutes by definition of $\varphi(q)$. The right quadrilateral commutes since q is H-equivariant. Thus the perimeter commutes, giving us that $\psi(\varphi(q)) = q \circ \mathrm{id} = q$. For the converse, let $p : X \to \mathrm{Ind}(Y)$. Then for $i \in I$ we have:

$$\begin{tikzcd}
 & \mathrm{Res}(X) = X \arrow[r, "\varphi(\psi(p))"] \arrow[d, "u_{g_i}^{-1}"] \arrow[ddl, "p"'] & \mathrm{Ind}(Y) \arrow[d, "\mathrm{pr}_i"] \\
 & X \arrow[r, "\psi(p)"] \arrow[d, "p"] & Y = Y_i \\
\mathrm{Ind}(Y) \arrow[r, "\mathrm{Ind}(v)_{g_i^{-1}}"] \arrow[d, "\mathrm{pr}_i"] & \mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_t"] \arrow[ur, "e"] & Y = Y_t \arrow[u, "v_{g_t}"'] \\
Y_i = Y \arrow[urr, "v_l"] & &
\end{tikzcd}$$

The top square commutes by definition of φ. The central triangle commutes by definition of ψ. The center right triangle commutes by definition of e. The left triangle commutes since p is a G-equivariant map. For the bottom triangle, we would like to invoke the definition of $\mathrm{Ind}(v)$ applied to g_i^{-1}. There is an $\ell \in I$ such that $t = j(g_i^{-1}, t)$. Then we have that $g_i^{-1} g_\ell = g_t h_{g_i^{-1},\ell}$. But since the right hand side is in H, g_i^{-1} and g_ℓ must be in the same coset. Thus $\ell = i$. Hence $1_G = g_i^{-1} g_i = g_t h_{g_i^{-1},\ell}$, so then $h_{g_i^{-1},\ell} = l$. So the bottom triangle indeed commutes by the definition of $\mathrm{Ind}(v)$. Then since $\mathrm{pr}_i \circ p = \mathrm{pr}_i \circ \varphi(\psi(p))$ for all $i \in I$, by the uniqueness of the universal property of products, $\varphi(\psi(p)) = p$. Hence ψ and φ are mutually inverse.

It remains to check naturality. For this, let $\alpha : (X', u') \to (X, u)$ and $\beta : (Y, v) \to (Y', v')$ be morphisms in the respective categories. Then we have the following diagram:

$$\begin{tikzcd}
X' = \mathrm{Res}(X') \arrow[r, "\mathrm{Res}(\alpha)"] \arrow[dd, "\alpha"] & \mathrm{Res}(X) \arrow[r, "\psi(p)"] \arrow[dd, "p"] & Y \arrow[r, "\beta"] & Y' \\
 & & Y = Y_t \arrow[r, "\beta"] \arrow[u, "u_{g_t}"'] & Y' = Y'_t \arrow[u, "u'_{g_t}"] \\
X \arrow[r, "p"] & \mathrm{Ind}(Y) \arrow[r, "\mathrm{Ind}(\beta)"] \arrow[uur, "e"] \arrow[ur, "\mathrm{pr}_t"'] & & \mathrm{Ind}(Y') \arrow[u, "\mathrm{pr}_t"] \arrow[uu, bend right, "e"']
\end{tikzcd}$$

The left square commutes trivially, and the center right, and right most triangles commute by definition of e. The center left triangle commutes by definition of $\psi(p)$. The upper right square commutes since β is H-equivariant, and the lower right square commutes by definition of $\mathrm{Ind}(\beta)$.

Thus the perimeter commutes. The top path is $[\mathrm{Hom}(\mathrm{Res}(\alpha),\beta)\circ\psi](p)$, and the bottom path is $[\psi\circ\mathrm{Hom}(\alpha,\mathrm{Ind}(\beta))](p)$. Thus the diagram commuting shows naturality, so we have an adjunction, as desired. □

Corollary 3.23. The internal hom of $\mathrm{Rep}_{\mathcal{C}}(H)$ as a $\mathrm{Rep}_{\mathcal{C}}(G)$-module category is given below:

$$\mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(H)}(X\otimes^H Y, Z)\cong \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(G)}(X,\mathrm{Ind}(Z\otimes Y^*))$$

Proof. Analogous to the proof of Corollary 3.17 using Theorem 3.22 and Corollary 2.5. □

We will now generalize Example 2.8 and Example 2.10 to being valued in a fusion category $\mathcal{C}$, which we assume to be strict, rather than Vec. Suppose that a finite group G has trivial action on a fusion category $\mathcal{C}$. Consider $\mathcal{C}$ as a module category over itself, and for $H\leq G$, consider H-equivariant structures on $\mathcal{C}$ such that $U_h=\mathrm{Id}_{\mathcal{C}}$ is trivial as a functor $\mathcal{C}\to\mathcal{C}=\mathcal{C}^h$ for all $h\in H$. Even though U_h might a priori not be trivial as a $\mathcal{C}$-module functor, the argument in Example 2.8 generalizes to show that in fact $(U_h,\tau_h)=\mathrm{Id}_{\mathcal{C}}$.

So the only thing with freedom (assuming that each U_h is trivial as a functor) are the natural isomorphisms $\mu_{h,k}:\mathrm{Id}_{\mathcal{C}}\cong\mathrm{Id}_{\mathcal{C}}$. By the fact that $\mu_{h,k}$ is a $\mathcal{C}$-module functor natural isomorphism, we get that $\mu_{h,k}(X\otimes Y)=\mathrm{id}_X\otimes\mu_{h,k}(Y):X\otimes Y\to X\otimes Y$. Thus by strictness we have

$$\mu_{h,k}(X)=\mu_{h,k}(X\otimes 1)=\mathrm{id}_X\otimes\mu_{h,k}(1)$$

So $\mu_{h,k}$ is determined by maps $\mu_{h,k}:1\xrightarrow{\sim}1$. Then by the compatibility condition we have that

$$\mu_{gh,k}(1)\circ\mu_{g,h}(1)=\mu_{g,hk}(1)\circ\mu_{h,k}(1)$$

Since $\mathcal{C}$ is a fusion category, we have that $\mathrm{Hom}_{\mathcal{C}}(1,1)\cong\Bbbk$. Notice that this implies that we have

$$\mu_{h,k}(X)=\mathrm{id}_X\otimes\mu_{h,k}(1)=\mu_{h,k}(1)\,\mathrm{id}_X$$

Where the last expression is in the $\Bbbk$-vector space $\mathrm{Hom}(X,X)$. Since $\mu_{h,k}$ is invertible, we can regard it as an element of $\Bbbk^\times$. Thus μ gives a function $\mu:H\times H\to\Bbbk^\times$ by $\mu(h,k)=\mu_{h,k}(1)$. This is a 2-cocycle by the above equation. Similarly to Example 2.10, such module categories are of the form below.

Definition 3.24. For a fusion category $\mathcal{C}$ with trivial action by G, $H \leq G$, let μ be a 2-cocycle on H, and define the category of *projective representations of H in $\mathcal{C}$ with Schur multiplier μ*, $\operatorname{Rep}_{\mathcal{C},\mu}(H)$, as follows: Let $\mathcal{M} = \mathcal{C}$ be a $\mathcal{C}$-module category in the obvious way, and give $\mathcal{M}$ a H-equivariant structure with U, τ trivial, and natural transformation defined by μ. Then $\operatorname{Rep}_{\mathcal{C},\mu}(H) := \mathcal{M}^H$, as in Definition 2.9.

In other words, $\operatorname{Rep}_{\mathcal{C},\mu}(H)$ has objects $(X, u = \{u_h : X \xrightarrow{\sim} X\}_{h\in H})$ with X and each u_h in $\mathcal{C}$ such that for $h, k \in H$, the following commutes:

$$\begin{tikzcd} X \arrow[r, "u_k"] \arrow[d, "\mu(h,k)\,\mathrm{id}"'] & X \arrow[d, "u_h"] \\ X \arrow[r, "u_{hk}"] & X \end{tikzcd}$$

And a morphism $f : (X, u) \to (Y, v)$ is a morphism in $\mathcal{C}$ such that for all $h \in H$ we have:

$$\begin{tikzcd} X \arrow[r, "f"] \arrow[d, "u_h"] & Y \arrow[d, "v_h"] \\ X \arrow[r, "f"] & Y \end{tikzcd}$$

We then have that $\operatorname{Rep}_{\mathcal{C},\mu}(H)$ is a $\operatorname{Rep}_{\mathcal{C}}(G)$-module category with the following action:

$$\otimes^{H,\mu} : \operatorname{Rep}_{\mathcal{C}}(G) \times \operatorname{Rep}_{\mathcal{C},\mu}(H); (X, u) \otimes^{H,\mu} (Y, v) := (\operatorname{Res}(X) \otimes Y, \operatorname{Res}(u) \otimes v)$$

This is easy to see that this is well defined since if $h \in H$, we have that

$$\begin{aligned}(\operatorname{Res}(u)_h \otimes v_h) \circ (\operatorname{Res}(u)_k \otimes v_k) =& (u_h \circ u_k) \otimes (v_h \otimes v_k) \\ =& u_{hk} \otimes (v_{hk} \circ \mu(h,k)\,\mathrm{id}) = u_{hk} \otimes (\mu(h,k) v_{hk}) \\ =& \mu(h,k)(v_{hk} \otimes u_{hk}) \\ =& (\operatorname{Res}(u)_{hk} \otimes v_{hk}) \circ \mu(h,k)\,\mathrm{id}_{\operatorname{Res}(X)\otimes Y}\end{aligned}$$

In order to compute internal homs for this action, i.e. an analogue of Theorem 3.19, we need an analogue of Lemma 3.18.

Lemma 3.25. Given $(X, u), (Y, v) \in \operatorname{Rep}_{\mathcal{C},\mu}(H)$, we have $(Y \otimes X^*, u \otimes (v^{-1})^*) \in \operatorname{Rep}_{\mathcal{C}}(H)$.

Proof. First notice that $((\mu_{h,k}(Y))^{-1})^* = ((\mu(h,k)\,\mathrm{id}_Y)^{-1})^* = (\mu(h,k)^{-1}\,\mathrm{id}_Y)^* = \mu(h,k)^{-1}\,\mathrm{id}_{Y^*}$ by the fact that composition is $\Bbbk$-linear. If we take the inverse of, and then the dual of the diagram that expresses that $(Y, v) \in \operatorname{Rep}_{\mathcal{C},\mu}(H)$ for $h, k \in H$, we get the following:

$$\begin{tikzcd} Y^* \arrow[r, "(v_k^{-1})^*"] \arrow[d, "\mu(h,k)^{-1}\,\mathrm{id}"'] & Y^* \arrow[d, "(v_h^{-1})^*"] \\ Y^* \arrow[r, "(v_{hk}^{-1})^*"] & Y^* \end{tikzcd}$$

Then tensoring that with the appropriate diagram for (X, u), we get the following:

$$\begin{tikzcd} X \otimes Y^* \arrow[r, "u_k \otimes (v_k^{-1})^*"] \arrow[d, "(\mu(h,k)\,\mathrm{id}) \otimes (\mu(h,k)^{-1}\,\mathrm{id})"'] & X \otimes Y^* \arrow[d, "u_h \otimes (v_h^{-1})^*"] \\ X \otimes Y^* \arrow[r, "u_{hk} \otimes (v_{hk}^{-1})^*"] & X \otimes Y^* \end{tikzcd}$$

Since $(\mu(h,k)\,\mathrm{id}) \otimes (\mu(h,k)^{-1}\,\mathrm{id}) = \mu(h,k)\mu(h,k)^{-1}\,\mathrm{id}_{X \otimes Y^*} = \mathrm{id}_{X \otimes Y^*}$, the above diagram expresses that $(Y \otimes X^*, u \otimes (v^{-1})^*) \in \mathrm{Rep}_{\mathcal{C}}(H)$, as desired. $\square$

Now we have an analogue of Theorem 3.19 for representations valued in a fusion category $\mathcal{C}$.

Theorem 3.26. *Let μ be as above, then we have an adjunction:*

$$\mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C},\mu}(H)}((U,\pi) \otimes^{H,\mu} (V,\rho), (W,\theta)) \cong \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(G)}((U,\pi), \mathrm{Ind}((W \otimes V^*, \theta \otimes (\rho^{-1})^*)))$$

In other words, $\mathrm{Ind}((W \otimes V^*, \theta \otimes (\rho^{-1})^*))$ *is the internal hom of* $\mathrm{Rep}_{\mathcal{C},\mu}(H)$ *as a* $\mathrm{Rep}_{\mathcal{C}}(G)$*-module.*

Proof. The proof is similar to the proof of Theorem 3.19. First we consider the case where $H = G$, here Res and Ind are trivial, so we want to show there is an adjunction:

$$\mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C},\mu}(H)}((U,\pi) \otimes^{H,\mu} (V,\rho), (W,\theta)) \cong \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(H)}((U,\pi), (W \otimes V^*, \theta \otimes (\rho^{-1})^*))$$

Since the morphisms in $\mathrm{Rep}_{\mathcal{C},\mu}(H)$ are defined the same way as how they are in $\mathrm{Rep}_{\mathcal{C}}(H)$, showing this is similar to the proof of Theorem 1.17 using Corollary 2.5. So then moving back to the general case, we have:

$$\begin{aligned} \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C},\mu}(H)}((U,\pi) \otimes^{H,\mu} (V,\rho), (W,\theta)) &= \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C},\mu}(H)}(\mathrm{Res}(U,\pi) \otimes (V,\rho), (W,\theta)) \\ &\cong \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(H)}(\mathrm{Res}(U,\pi), (W \otimes V^*, \theta \otimes (\rho^{-1})^*)) \\ &\cong \mathrm{Hom}_{\mathrm{Rep}_{\mathcal{C}}(G)}((U,\pi), \mathrm{Ind}((W \otimes V^*, \theta \otimes (\rho^{-1})^*))) \end{aligned}$$

The first isomorphism is provided by the $H = G$ case applied to $\mathrm{Res}(U, \pi)$. As remarked after Theorem 3.19, on the right hand side of the first equallity, we use the bare $\otimes$ to denote the action of $\mathrm{Rep}_{\mathcal{C}}(H)$ on $\mathrm{Rep}_{\mathcal{C},\mu}(H)$. □

3.5 Equivariantizations of Module Categories using the Group Action

Consider a fusion category $\mathcal{C}$ with action T on $\mathcal{C}$ by a finite group G. Let $H \leq G$ and consider $\mathcal{M} = \mathcal{C}$ as a $\mathcal{C}$-module category with the obvious action. Let us attempt to give this an H-equivariant $\mathcal{C}$-module structure with U coming from T.

Proposition 3.27. We have that $\mathcal{C}$ is an H-equivariant $\mathcal{C}$-module category with $(U_h, \tau_h) = (T_h, t_h^{-1})$ and $\mu_{h,k} = \gamma_{h,k}$ if γ satisfies the following "2-cocycle condition" for all X:

$$\gamma_{hk,l}(X) \circ \gamma_{h,k}(T_l(X)) = \gamma_{h,kl}(X) \circ T_h(\gamma_{k,l}(X))$$

Proof. We clearly have that $U_h : \mathcal{M} \to \mathcal{M}^h$ since it's defined as

$$T_h(X \otimes Y) \xrightarrow{t_h^{-1}(X,Y)} T_h(X) \otimes T_h(Y) = X \otimes^h T_h(Y)$$

Then the diagrams expressing that T_h is a monoidal functor are exactly the diagrams expressing that it is a $\mathcal{C}$-module functor. That $\mu_{g,h}$ is a $\mathcal{C}$-module transformation similarly follows immediately from the fact that $\gamma_{h,k}$ is a monoidal natural transformation. The restriction on γ as above is exactly the compatibility condition in Definition 2.7. □

We will now assume that we have $\mathcal{C}, H, G, \gamma$ as above, and consider the equivariantization (as a module category), $\mathcal{C}^H$, of $\mathcal{M}$, which agrees with the equivariantization of $\mathcal{C}$ as a fusion category with action by H given by restricting the action from G. There is an obvious restriction functor:

$$\mathrm{Res} : \mathcal{C}^G \to \mathcal{C}^H ; (X, \{u_g\}_{g \in G}) \mapsto (X, \{u_h\}_{h \in H})$$

We then get that $\mathcal{C}^H$ is a $\mathcal{C}^G$ module category in the obvious way with

$$(X, u) \otimes^H (Y, v) := \mathrm{Res}(X, u) \otimes (Y, v) = (X \otimes Y, \{[u \otimes v]_h := (u_h \otimes v_h) \otimes t_h^{-1}(X, Y)\})$$

As before, let $H \leq G$ and again choose representatives $\{g_i\}_{i\in I}$ for the cosets G/H and define $gg_i = g_{j(g,i)}h_{g,i}$, $X_i := X$, and $u_g^i : T_{h_{g,i}}(X_i) \to X_{j(g,i)}$ by $u_{h_{g,i}}$. We define an induction functor by:

$$\mathrm{Ind} : \mathcal{C}^H \to \mathcal{C}^G; (X,u) \mapsto (\mathrm{Ind}(X) := \bigoplus_i T_{g_i}(X_i), \{\mathrm{Ind}(u)_g\}_g)$$

Where $\mathrm{Ind}(u)_g$ is defined as follows, and the first isomorphism is given by Remark 1.23:

$$\begin{aligned}
\mathrm{Ind}(u)_g : T_g(\mathrm{Ind}(X)) =& T_g(\bigoplus_i T_{g_i}(X_i)) \\
\cong& \bigoplus_i T_g(T_{g_i}(X_i)) \\
\xrightarrow{\bigoplus_i \gamma_{g,g_i}(X_i)}& \bigoplus_i T_{gg_i}(X_i) \\
=& \bigoplus_i T_{g_{j(g,i)}h_{g,i}}(X_i) \\
\xrightarrow{\bigoplus_i \gamma^{-1}_{g_{j(g,i)},h_{g,i}}(X_i)}& \bigoplus_i T_{g_{j(g,i)}}(T_{h_{g,i}}(X_i)) \\
\xrightarrow{\bigoplus_i T_{g_{j(g,i)}}(u_g^i)}& \bigoplus_i T_{g_{j(g,i)}}(X_{j(g,i)}) = \mathrm{Ind}(X)
\end{aligned}$$

Let pr_i again be projection maps. Using the universal property, we define $\mathrm{Ind}(u)_g$ more precisely by using:

$$\begin{tikzcd}
T_g(\mathrm{Ind}(X)) \arrow[r, "T_g(\mathrm{pr}_\ell)"] \arrow[dddd, dashed, "\mathrm{Ind}(u)_g"] & T_g(T_{g_\ell}(X_\ell)) \arrow[d, "\gamma_{g,g_\ell}(X_\ell)"] \\
& T_{gg_\ell}(X_\ell) \arrow[d, "\gamma^{-1}_{g_{j(g,\ell)},h_{g,\ell}}(X_\ell)"] \\
& T_{g_{j(g,\ell)}}(T_{h_{g,\ell}}(X_\ell)) \arrow[dd, "T_{g_{j(g,\ell)}}(u_g^\ell)"] \\
& \\
\mathrm{Ind}(X) \arrow[r, "\mathrm{pr}_{j(g,\ell))}"] & T_{g_{j(g,\ell)}}(X_{j(g,\ell)})
\end{tikzcd}$$

Notice that this is a direct generalization of the functor in Lemma 4.6 of Drinfeld et al. (2010). Similarly to section 3.4, it suffices to check that this satisfies the desired commutativity on each component via the universal property of the product. By equivariance of (X,u) we have that the following commutes:

$$\begin{tikzcd}
T_{h_{k,j(g,\ell)}}(T_{h_{g,\ell}}(X_\ell)) \arrow[r, "T_{h_{k,j(g,\ell)}}(u_{h_{g,\ell}})"] \arrow[d, "\gamma_{h_{k,j(g,\ell)},h_{g,\ell}}(X_\ell)"'] & T_{h_{k,j(g,\ell)}}(X_{j(g,\ell)}) \arrow[d, "u_{h_{k,j(g,\ell)}}"] \\
T_{h_{kg,i}}(X_\ell) \arrow[r, "u_{h_{kh,\ell}}"] & X_{j(kg,\ell)}
\end{tikzcd}$$

Now refer to the diagram on the following page. The rectangle of parallel arrows commute clearly since $gg_\ell = g_{j(g,\ell)}h_{g,\ell}$. The leftmost rectangle, and the two quadrilaterals in the center all commute via the 2-cocycle condition for γ. The top two rightmost squares commute via naturality of γ. Finally, the bottom right square is just the diagram for equivariance of (X,u) we drew above sent through $T_{g_{j(kg,\ell)}}$. Hence the above commutes and $\mathrm{Ind}(u)$ is an equivariant structure for $\mathrm{Ind}(X)$.

Given an equivariant map $f:(X,u)\to(Y,v)$, we define $\mathrm{Ind}(f)$ as below. That this is an equivariant map follows similarly to the case in section 3.4, but using naturality of γ as well as that f is equivariant, and hence Ind is well defined as a functor.

$$\begin{tikzcd}
\mathrm{Ind}(X) \arrow[r, "\mathrm{pr}_\ell"] \arrow[d, dashed, "\mathrm{Ind}(f)"] & T_{g_\ell}(X_\ell) \arrow[d, "T_{g_\ell}(f)"] \\
\mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_\ell"] & T_{g_\ell}(Y_\ell)
\end{tikzcd}$$

Proposition 3.28. (Generalized Frobenius Reciprocity) Let $(X,u)\in\mathcal{C}^G$ and $(Y,v)\in\mathcal{C}^H$, then we get an adjunction:

$$\mathrm{Hom}_{\mathcal{C}^H}(\mathrm{Res}(X),Y)\cong\mathrm{Hom}_{\mathcal{C}^G}(X,\mathrm{Ind}(Y))$$

Proof. The proof will be similar to the proof of Theorem 3.22, but we will still go through it, albeit referring to the proof of Theorem 3.22 when the details are essentially the same. We start out by again decomposing $1_G = g_t l$ with $l\in H$ and defining an evaluation map:

$$e:\mathrm{Ind}(Y)\xrightarrow{\mathrm{pr}_t}T_{g_t}(Y_t)=T_{g_t}(Y)\xrightarrow{v_{g_t}}Y$$

$$\begin{tikzcd}
T_k(T_g(T_{g_\ell}(X_\ell))) \arrow[r, "T_k(\gamma_{g,g_\ell}(X_\ell))"] \arrow[ddd, "\gamma_{k,g}(T_{g_\ell}(X_\ell))"] & T_k(T_{gg_\ell}(X_\ell)) \arrow[r, equal] \arrow[ddd, "\gamma_{k,gg_\ell}(X_\ell)"] & T_k(T_{g_{j(g,\ell)}h_{g,\ell}}(X_\ell)) \arrow[r, "T_k(\gamma^{-1}_{g_{j(g,\ell)},h_{g,\ell}}(X_\ell))"] \arrow[ddd, "\gamma_{k,g_{j(g,\ell)}h_{g,\ell}}"] & T_k(T_{g_{j(g,\ell)}}(T_{h_{g,\ell}}(X_\ell))) \arrow[r, "T_k(T_{g_{j(g,\ell)}}(u_{h_{g,\ell}}))"] \arrow[d, "\gamma_{k,g_{j(g,\ell)}}(T_{h_{g,\ell}}(X_\ell))"] & T_k(T_{g_{j(g,\ell)}}(X_{j(g,\ell)})) \arrow[d, "\gamma_{k,g_{j(g,\ell)}}(X_{j(g,\ell)})"] \\
 & & & T_{kg_{j(g,\ell)}}(T_{h_{g,l}}(X_\ell)) \arrow[r, "T_{kg_{j(g,\ell)}}(u_{h_{g,\ell}})"] \arrow[d, "\gamma^{-1}_{g_{j(kg,\ell)},h_{k,j(g,\ell)}}(T_{h_{g,\ell}}(X_\ell))"] \arrow[ddl, "\gamma_{kg_{j(g,\ell)},h_{g,\ell}}(X_\ell)"'] & T_{kg_{j(g,\ell)}}(X_{j(g,\ell)}) \arrow[d, "\gamma^{-1}_{g_{j(kg,\ell)},h_{k,j(g,\ell)}}(X_{j(g,\ell)})"] \\
 & & & T_{g_{j(kg,\ell)}}(T_{h_{k,j(g,\ell)}}(T_{h_{g,\ell}}(X_\ell))) \arrow[r, "T_{g_{j(kg,\ell)}}(T_{h_{k,j(g,\ell)}}(u_{h_{g,\ell}}))"] \arrow[d, "T_{g_{j(kg,\ell)}}(\gamma_{h_{k,j(g,\ell)}}(X_\ell))"] & T_{g_{j(kg,\ell)}}(T_{h_{k,j(g,\ell)}}(X_{j(g,\ell)})) \arrow[d, "T_{g_{j(kg,\ell)}}(u_{h_{k,j(g,\ell)}})"] \\
T_{kg}(T_{g_\ell}(X_\ell)) \arrow[r, "\gamma_{kg,g_\ell}(X_\ell)"] & T_{(kg),g_\ell}(X_\ell) \arrow[r, equal] & T_{g_{j(kg,\ell)}h_{kh,\ell}}(X_\ell) \arrow[r, "\gamma^{-1}_{g_{j(kg,\ell)},h_{kg,\ell}}(X_\ell)"] & T_{g_{j(kg,\ell)}}(T_{h_{kg,\ell}}(X_\ell)) \arrow[r, "T_{g_{j(kg,\ell)}}(u_{h_{kg,\ell}})"] & T_{g_{j(kg,\ell)}}(X_{j(kg,\ell)})
\end{tikzcd}$$

We claim this is H-equivariant, which can be seen from the below diagram whose left polygon commutes by definition of $\mathrm{Ind}(u)$, and whose right two quadrilaterals commute by equivariance:

$$\begin{tikzcd}
T_h(\mathrm{Ind}(Y)) \arrow[r, "T_h(\mathrm{pr}_t)"] \arrow[ddd, "\mathrm{Ind}(u)_h"] & T_h(T_{g_t}(Y_t)) \arrow[r, "T_h(v_{g_t})"] \arrow[d, "\gamma_{h,g_t}(Y_t)"] & T_h(Y_t) = T_h(Y) \arrow[ddd, "v_h"] \\
& T_{hg_t}(Y_t) = T_{g_t(lhg_t)}(Y_t) \arrow[d, "\gamma^{-1}_{g_t,(lhg_t)}(Y_t)"'] \arrow[ddr, "v_{hg_t}=v_{g_t(lhg_t)}"] & \\
& T_{g_t}(T_{lhg_t}(Y_t)) \arrow[d, "T_{g_t}(v_{lhg_t})"] & \\
\mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_t"] & T_{g_t}(Y_t) = T_{g_t}(Y) \arrow[r, "v_{g_t}"] & Y
\end{tikzcd}$$

We define a map $\psi : \mathrm{Hom}_{\mathcal{C}^G}(X, \mathrm{Ind}(Y)) \to \mathrm{Hom}_{\mathcal{C}^H}(\mathrm{Res}(X), Y)$ by $(p : X \to \mathrm{Ind}(Y)) \mapsto e \circ p$ which will clearly be an H-equivariant map since both e and p are. Conversely, we define a map $\varphi : \mathrm{Hom}_{\mathcal{C}^H}(\mathrm{Res}(X), Y) \cong \mathrm{Hom}_{\mathcal{C}^G}(X, \mathrm{Ind}(Y))$ as follows: given $q : \mathrm{Res}(X) \to Y$ we use the universal property of the product to get a unique map $\varphi(q)$:

$$\begin{tikzcd}
X \arrow[r, "u_{g_i}^{-1}"] \arrow[d, dashed, "\varphi(q)"] & T_{g_i}(X) = T_{g_i}(\mathrm{Res}(X)) \arrow[d, "T_{g_i}(q)"] \\
\mathrm{Ind}(Y) \arrow[r, "\mathrm{pr}_i"] & T_{g_i}(Y_i) = T_{g_i}(Y)
\end{tikzcd}$$

To see that this is a G-equivariant map is similar to the process in Theorem 3.22, now we just also need to use the naturality of γ. Showing that φ and ψ are inverses as well as naturality is essentially identical to before, just with some added Ts. $\square$

Corollary 3.29. The internal hom of the action of $\mathcal{C}^G$ on $\mathcal{C}^H$ is given below:

$$\mathrm{Hom}_{\mathcal{C}^H}(X \otimes^H Y, Z) \cong \mathrm{Hom}_{\mathcal{C}^G}(X, \mathrm{Ind}(Z \otimes Y^*))$$

Proof. A clear generalization of the proof of Corollary 3.23. $\square$

Proposition 3.30. We have that $\mathcal{C}$ is an H-equivariant $\mathcal{C}$-module category with $(U_h, \tau_h) = (T_h, t_h^{-1})$ and $\mu_{h,k} = \lambda(h,k)\gamma_{h,k}$ where λ is a 2-cocycle and γ satisfies the 2-cocycle condition as in Proposition 3.27.

Proof. As before, T_h being a monoidal functor immediately gives that it is a $\mathcal{C}$-module functor. The fact that $\mu_{h,k}$ is a $\mathcal{C}$-module natural transformation and satisfies the compatibility condition follows immediately from the $\Bbbk$-linearity of composition as well as the fact that $\gamma_{h,k}$ is a monoidal natural transformation and satisfies the 2-cocycle condition. $\square$

To avoid confusion, in this case we will denote the equivariantization of $\mathcal{C}$ as a module category as $\mathcal{C}^{H,\lambda}$. Clearly when λ is trivial, we recover the familiar equivariantization as a fusion category. If T is trivial then we recover $\mathrm{Rep}_{\mathcal{C},\lambda}(H)$.

Lemma 3.31. Given $(X,u),(Y,v) \in \mathcal{C}^{H,\lambda}$, we have that $(Y \otimes X^*, u \otimes (v^{-1})^*) \in \mathcal{C}^H$.

Proof. Using the $\Bbbk$-linearity of composition, the proof follows from Remark 2.4 and Lemma 3.25. $\square$

Theorem 3.32. *The internal hom of the action of $\mathcal{C}^G$ on $\mathcal{C}^{H,\lambda}$ is given below:*

$$\mathrm{Hom}_{\mathcal{C}^{H,\lambda}}(X \otimes^H Y, Z) \cong \mathrm{Hom}_{\mathcal{C}^G}(X, \mathrm{Ind}(Z \otimes Y^*))$$

Proof. A clear generalization of Theorem 3.26. $\square$

CHAPTER 4. CONCLUSION

In this thesis we computed the internal homs with respect to a module category action in a few key cases. We focused on module categories over equivariantizations since the important example of categories of representations arise in this case. Our results clearly line up with the known results and the intuition that we should be inducing an internal hom. We started with the very modest assumptions that we were equivariantizating the category of finite dimensional vector spaces, then to arbitrary fusion categories with trivial group action, and then to arbitrary fusion categories where the group action satisfies a 2-cocycle condition. In each generalization, the procedure was the same. We started by assuming that the equivariant module category structure was trivial, proved a version of Frobenius reciprocity to get the internal homs in this case, and then used this easier case to prove it when the equivariant module category structure was non trivial (potentially with some assumptions).

This work will allow for future researchers to have a more concrete hold on module categories over equivariantizations of fusion categories. Not only by clearing up the classification theorem, but by computing internal homs, which can give a more explicit form for internal algebras for which module categories are equivalent to the category of modules.

While we were only able to compute these internal homs with respect to the equivariantized case of a fusion category acting on itself and then using the group action to get an equivariant module category structure, an area for future research may be to generalize this. This work was a large generalization of the known case for the category of finite dimensional vector spaces, but the general cases where the module category and the equivariant module category structure may differ is still unknown.

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